United States Patent	12401877
Kind Code	B2
Date of Patent	August 26, 2025
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Camera module with improved mobility and camera apparatus comprising same

Abstract

A camera module includes a first driving part on a lower surface of an image sensor substrate, and a second driving part on a lower surface of the housing to face a lower surface of the first driving part. The first driving part includes a first driving part of a first group configured to move the image sensor substrate in a first direction with respect to the lens barrel, and a second driving part of a second group configured to move the image sensor substrate in a second direction intersecting the first direction with respect to the lens barrel. The second driving part includes a second driving part of a first group disposed overlapping the first driving part of the first group in a vertical direction, and a second driving part of a second group disposed overlapping the first driving part of the second group in the vertical direction.

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Appl. No.: **18/627126**

Filed: **April 04, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20240251149 A1	Jul. 25, 2024

Foreign Application Priority Data

KR	10-2019-0002244	Jan. 08, 2019

Related U.S. Application Data

continuation parent-doc US 17420433 US 11979652 WO PCT/KR2020/000316 20200108 child-doc US 18627126

Publication Classification

Int. Cl.: **H04N23/55** (20230101); **H04N23/51** (20230101); **H04N23/54** (20230101); **H04N23/57** (20230101)

U.S. Cl.:

CPC **H04N23/55** (20230101); **H04N23/51** (20230101); **H04N23/54** (20230101); **H04N23/57** (20230101);

Field of Classification Search

CPC: H04N (23/55); H04N (23/51); H04N (23/54); H04N (23/57); H04N (23/687); H05K (1/147); H05K (1/189); H05K (2201/10121); H05K (2201/10151); H05K (1/111); H05K (1/181)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/420,433, filed on Jul. 2, 2021, which is the National Phase of PCT International Application No. PCT/KR2020/000316, filed on Jan. 8, 2020, which claims priority under 35 U.S.C. 119(a) to Patent Application No. 10-2019-0002244, filed in the Republic of Korea on Jan. 8, 2019, all of which are hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD
(1) The embodiment relates to a camera module, and more particularly, to a camera module for relatively moving an image sensor with respect to a lens barrel, and a camera device including the same.
BACKGROUND ART
(2) In general, camera devices are mounted in mobile communication terminals and portable devices such as MP3 players, as well as electronic devices such as automobiles, endoscopes, and CCTVs. Such camera devices are gradually being developed centering on high pixels, and miniaturization and thinning are in progress. In addition, current camera devices are being changed so that various additional functions can be implemented at low manufacturing cost.
(3) The camera device as described above includes a lens barrel accommodating a lens, a lens holder coupled to the lens barrel, an image sensor disposed in the lens holder, and a driving substrate on which the image sensor is mounted. At this time, the lens prod provides an image signal of a subject to the image sensor. And the image sensor converts the image signal into an electrical signal.
(4) Here, the accuracy of the image signal in the camera device is determined according to a focal length defined as the distance between the lens and the image sensor.
(5) Accordingly, the camera device provides focus compensation or shake compensation by moving the lens barrel relative to the image sensor. That is, the camera device moved the lens barrel accommodating the lens relative to the image sensor in the X-axis, Y-axis, and Z-axis. At this time, the camera device required an elastic member such as at least six springs to move the lens barrel. In addition, the elastic member was bonded to the lens barrel by bonding.
(6) However, the camera device according to the prior art as described above, by moving the lens barrel, the upper spring plate disposed on the lens barrel, the lower spring plate under the lens

barrel, and an elastic wire for fixing the Z-axis (elastic wire) Includes structures such as Accordingly, the module structure of the camera device according to the prior art has a complicated problem.
(7) In addition, the camera device according to the prior art requires a plurality of elastic members for moving the lens barrel, and there is a problem in that the number of assembling steps of the plurality of elastic members increases.

DISCLOSURE

Technical Problem

(8) In the embodiment, it is possible to provide a substrate for an image sensor having a new structure and a camera module including the same.
(9) In addition, in the embodiment, it is possible to provide an image sensor substrate for an image sensor moving relative to a lens barrel, and a camera module including the same.
(10) In addition, in the embodiment, a substrate for an image sensor capable of not only movement of the X-axis, Y-axis, and Z-axis, but also tilt correction, and a camera module including the same are provided.
(11) In addition, in the embodiment, a substrate for an image sensor capable of simplifying a spring structure for providing an auto focus function or a camera shake compensation function, and a camera module including the same can be provided.
(12) The technical problems to be achieved in the proposed embodiment are not limited to the technical problems mentioned above, and other technical problems not mentioned may be clearly understood by those of ordinary skill in the art to which the proposed embodiment belongs from the following description.

Technical Solution

(13) A camera module according to an embodiment includes a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; an image sensor disposed on the image sensor substrate; a first driving part disposed under a lower surface of the image sensor substrate; and a second driving part disposed under a lower surface of the housing and having an upper surface facing a lower surface of the first driving part, wherein the image sensor substrate includes: a substrate portion; a conductive pattern portion disposed on the substrate portion; and a support layer disposed under the substrate portion, wherein the first driving part includes: a first driving part of a first group configured to move the image sensor substrate in a first direction with respect to the lens barrel; and a second driving part of a second group configured to move the image sensor substrate in a second direction intersecting the first direction with respect to the lens barrel, wherein the second driving part includes: a second driving part of a first group disposed overlapping the first driving part of the first group in a vertical direction; and a second driving part of a second group disposed overlapping the first driving part of the second group in the vertical direction.
(14) In addition, the first driving part includes a permanent magnet, and the second driving part includes a coil.
(15) In addition, the first driving part of the first group includes a first-first driving part and a first-third driving part respectively disposed under a lower surface of the spring plate and having a N pole and a S pole arranged in the first direction, and the first driving part of the second group includes a first-second driving part and first-fourth driving part respectively disposed on the lower surface of the spring plate and having a N pole and a S pole arranged in the second direction.
(16) In addition, arrangement directions of the N poles and S poles of the first-first to first-fourth driving parts are different from each other.
(17) In addition, the first direction is an X-axis direction, the second direction is a Y-axis direction, wherein the N poles and the S poles of the first-first driving part and the first-third driving part are arranged in a +X-axis direction and a −X-axis direction opposite to each other, and wherein the N poles and S poles of the first-second driving part and the first-fourth driving part are arranged in a

+Y axis direction and a −Y axis direction opposite to each other.
(18) In addition, the arrangement direction of the N pole and S pole of each of the first-first to first-fourth driving parts is perpendicular to the arrangement direction of an adjacent driving part.
(19) In addition, the first-first driving part and the first-third driving part are respectively disposed in corner regions in a diagonal direction to each other under the lower surface of the substrate portion, and the first-second driving part and the first-fourth driving part are respectively disposed in corner regions in a diagonal direction to each other.
(20) In addition, the first driving part of the first group is configured to move the image sensor substrate in the +X-axis direction or the −X-axis direction when currents in different directions are applied to the first-first driving part and the first-third driving part, and the first driving part of the first group is configured to tilt the image sensor substrate when currents in the same direction are applied to the first-first driving part and the first-third driving part.
(21) In addition, the first driving part of the second group is configured to move the image sensor substrate in the +Y-axis direction or the −Y-axis direction when currents in different directions are applied to the first-second driving part and the first-fourth driving part, and the first driving part of the second group is configured to tilt the image sensor substrate when currents in the same direction are applied to the first-second driving part and the first-fourth driving part.
(22) In addition, the camera module includes a third driving part disposed on the support layer so that a side surface overlaps the first driving part of the first group and the first driving part of the second group in a horizontal direction, and configured to move the image sensor substrate in a third direction corresponding to a Z-axis.
(23) In addition, the camera module includes a third driving part disposed under on the bottom surface of the housing and overlapping the first driving part in a vertical direction, wherein an upper surface of the third driving part includes a first region overlapping the S pole of the first-first driving part in a vertical direction; a second region overlapping the S pole of the first-second driving part in a vertical direction; a third region overlapping the S pole of the first-third driving part in a vertical direction; and a fourth region overlapping the S pole of the first-fourth driving part in a vertical direction, and wherein the second driving part overlaps with a remaining region of lower surfaces of the first-first driving part to the first-fourth driving part except for the region overlapping the third driving part.
(24) In addition, the substrate portion may include a spring plate having an elastic member disposed in a first open region; and an insulating layer disposed on the spring plate and having a second open region exposing the first open region, wherein the spring plate includes a first plate part; and a second plate part disposed around the first plate part with the first open region interposed therebetween, and connected to the first plate part by the elastic member, wherein the insulating layer includes a first insulating part disposed on the first plate part; a second insulating part disposed on the second plate part; wherein the conductive pattern portion includes a first lead pattern part disposed on the first insulating part, a second lead pattern part disposed on the second insulating part; and an extension pattern part disposed between the first and second lead pattern parts and floating on the second open region; and wherein the first driving part is disposed under a lower surface of the first plate part.
(25) In addition, the substrate portion includes an insulating layer; and a bonding sheet disposed on the insulating layer, wherein the insulating layer includes a first insulating part; a second insulating part disposed around the first insulating part and spaced apart from the first insulating part with a first open region therebetween; and an extension insulating part disposed in the first open region and connecting the first insulating part and the second insulating part; wherein the conductive pattern portion includes a first lead pattern part disposed on the first insulating part; a second lead pattern part disposed on the second insulating part; and an extension pattern part disposed on the extension insulating part and connecting between the first lead pattern part and the second lead pattern part, wherein the extension insulating part and the extension pattern part have a spring

shape, and wherein the first driving part is disposed under a lower surface of the first insulating part.

(26) On the other hand, a camera device according to an embodiment includes a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; a first driving part disposed under a lower surface of the image sensor substrate; a second driving part disposed under a lower surface of the housing and having an upper surface facing a lower surface of the first driving part; a third driving part disposed on a side surface of the image sensor substrate and having a side surface facing a side surface of the first driving part; an image sensor disposed on the image sensor substrate; and a flexible circuit board including a first connector part disposed in the housing and electrically connected to the image sensor substrate, a second connector part disposed outside the housing and electrically connected to an external device, and a connection part connecting the first and second connector parts, wherein the image sensor substrate includes: a substrate portion; a conductive pattern portion disposed on the substrate portion; and a support layer disposed under the substrate portion, wherein the first driving part includes: a first driving part of a first group configured to move the image sensor substrate in a first direction with respect to the lens barrel; and a second driving part of a second group configured to move the image sensor substrate in a second direction intersecting the first direction with respect to the lens barrel, wherein the second driving part includes: a second driving part of a first group disposed overlapping the first driving part of the first group in a vertical direction; and a second driving part of a second group disposed overlapping the first driving part of the second group in the vertical direction, wherein the first driving part of the first group includes a first-first driving part and a first-third driving part respectively disposed under a lower surface of the spring plate and having a N pole and a S pole arranged in the first direction, and the first driving part of the second group includes a first-second driving part and first-fourth driving part respectively disposed on the lower surface of the spring plate and having a N pole and a S pole arranged in the second direction, and wherein arrangement directions of the N poles and S poles of the first-first to first-fourth driving parts are different from each other.

(27) In addition, the first-first driving part and the first-third driving part are respectively disposed in corner regions in a diagonal direction to each other under the lower surface of the substrate portion, and the first-second driving part and the first-fourth driving part are respectively disposed in corner regions in a diagonal direction to each other, wherein the first driving part of the first group is configured to move the image sensor substrate in the +X-axis direction or the −X-axis direction when currents in different directions are applied to the first-first driving part and the first-third driving part, and the first driving part of the first group is configured to tilt the image sensor substrate when currents in the same direction are applied to the first-first driving part and the first-third driving part, wherein the first driving part of the second group is configured to move the image sensor substrate in the +Y-axis direction or the −Y-axis direction when currents in different directions are applied to the first-second driving part and the first-fourth driving part, and the first driving part of the second group is configured to tilt the image sensor substrate when currents in the same direction are applied to the first-second driving part and the first-fourth driving part.

Effects of the Invention

(28) According to an embodiment, in order to implement the OIS and AF functions of the camera module, instead of moving the conventional lens barrel, the image sensor is moved with respect to the lens barrel in the X-axis, Y-axis and Z-axis directions. Accordingly, the camera module according to the embodiment may remove a complex spring structure for implementing the OIS and AF functions, and thus the structure may be simplified. In addition, by moving the image sensor according to the embodiment with respect to the lens barrel, it is possible to provide a stable structure compared to the conventional structure.

(29) In addition, according to an embodiment, the extension pattern part electrically connected to the image sensor has a spring structure and is disposed to float on a spring plate. Accordingly, the

camera module may move the image sensor while elastically supporting the image sensor more stably. In addition, the length of the extension pattern part is at least 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part. Accordingly, it is possible to minimize noise generation while improving the mobility of the image sensor substrate.
(30) In addition, according to an embodiment, by preventing the elastic member and the extension pattern part from being aligned with each other in the vertical direction, it is possible to solve an electrical reliability problem that may occur due to a contact between the elastic member and the extension pattern part.
(31) In addition, in the insulating layer in the embodiment, an extension insulating part having a spring shape is disposed in a region that vertically overlaps with the extension pattern part, so that the spring plate can be removed. Accordingly, the camera module may move the image sensor with respect to the lens barrel while elastically supporting the image sensor more stably, and may minimize the product volume.
(32) In addition, in the embodiment, the width of the extension insulating part is greater than the width of the extension pattern part, so that the extension pattern part can be stably supported by the extension insulating part, and thus operation reliability can be improved.
(33) In addition, in the embodiment, the first driving part composed of a permanent magnet is disposed so that the same poles are adjacent to each other with respect to the center, while the N poles and the S poles of the permanent magnets adjacent to each other are perpendicular to each other. According to this, in the embodiment, it is possible to secure the arrangement freedom of the second driving part and the third driving part for adjusting the electromagnetic force in response to the permanent magnet, it is possible to move in the X-axis and Y-axis, and stable tilt control is also possible.

Description

DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a view showing a camera module according to a comparative example.
(2) FIG. **2** is a view showing a camera device according to an embodiment.
(3) FIG. **3** is a view showing a spring plate shown in FIG. **2**.
(4) FIG. **4** is a view showing first and second bonding sheets shown in FIG. **2**.
(5) FIG. **5** is a view showing an insulating layer according to an embodiment.
(6) FIG. **6** is a view showing a conductive pattern portion according to an embodiment.
(7) FIG. **7** is a plan view of an image sensor substrate according to an embodiment.
(8) FIG. **8** is a view showing an arrangement structure of a first driving part according to an embodiment.
(9) FIG. **9** is a view showing an arrangement structure of a second driving part according to an embodiment.
(10) FIG. **10** is a view showing an arrangement structure of a third driving part according to an embodiment.
(11) FIG. **11** is a view showing an arrangement structure of a third driving part according to another embodiment.
(12) FIG. **12** is a view showing an arrangement structure of a third driving part according to another embodiment.
(13) FIG. **13** is a view showing illustrating a connection structure between a flexible circuit board and an image sensor substrate according to an embodiment.
(14) FIG. **14** is a view for explaining in detail a layer structure of a conductive pattern portion according to an embodiment.
(15) FIG. **15** is a view showing a camera module according to another embodiment.

(16) FIG. **16** is a view showing a camera device according to another exemplary embodiment.
(17) FIG. **17** is a view showing an insulating layer shown in FIG. **16**.
(18) FIG. **18** is a view showing a second bonding sheet shown in FIG. **16**.
(19) FIG. **19** is a view showing a conductive pattern portion according to an embodiment.
(20) FIG. **20** is a plan view of an image sensor substrate according to another embodiment.
(21) FIG. **21** is a view showing an arrangement structure of a first driving part according to another embodiment.

BEST MODE

(22) Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings.

(23) However, the spirit and scope of the present invention is not limited to a part of the embodiments described, and may be implemented in various other forms, and within the spirit and scope of the present invention, one or more of the elements of the embodiments may be selectively combined and replaced.

(24) In addition, unless expressly otherwise defined and described, the terms used in the embodiments of the present invention (including technical and scientific terms may be construed the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs, and the terms such as those defined in commonly used dictionaries may be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art. Further, the terms used in the embodiments of the present invention are for describing the embodiments and are not intended to limit the present invention.

(25) In this specification, the singular forms may also include the plural forms unless specifically stated in the phrase, and may include at least one of all combinations that may be combined in A, B, and C when described in “at least one (or more) of A (and), B, and C”. Further, in describing the elements of the embodiments of the present invention, the terms such as first, second, A, B, (a), and (b) may be used.

(26) These terms are only used to distinguish the elements from other elements, and the terms are not limited to the essence, order, or order of the elements. In addition, when an element is described as being “connected”, “coupled”, or “connected” to another element, it may include not only when the element is directly “connected” to, “coupled” to, or “connected” to other elements, but also when the element is “connected”, “coupled”, or “connected” by another element between the element and other elements.

(27) In addition, when described as being formed or disposed “on (over)” or “under (below)” of each element, the “on (over)” or “under (below)” may include not only when two elements are directly connected to each other, but also when one or more other elements are formed or disposed between two elements. Further, when expressed as “on (over)” or “under (below)”, it may include not only the upper direction but also the lower direction based on one element.

(28) Hereinafter, embodiments of the present disclosure will be described in detail with reference to the accompanying drawings.

(29) FIG. **1** is a view showing a camera module according to a comparative example.

(30) A camera module having an optical image stabilizer (OIS) function and an Auto Focusing (AF) function requires at least two spring plates.

(31) The camera module according to the comparative example may have two spring plates. The camera module according to the comparative example requires an elastic member such as at least six springs for the spring plate.

(32) Referring to FIG. **1**, the camera module according to the comparative example includes an optical system including a lens assembly, an infrared cut-off filter, and a sensor unit. That is, the camera module according to the comparative example includes a lens barrel **10**, a lens assembly **20**, a first elastic member **31**, a second elastic member **32**, a first housing **41**, a housing **42**, an infrared cut-off filter **50**, a sensor unit **60**, a circuit board **80**, and driving parts **71**, **72**, **73**, and **74**.

(33) In this case, the lens barrel **10** is connected to the first housing **41**. That is, the lens barrel **10** is connected to the first housing **41** via the first elastic member **31**. That is, the lens barrel **10** is connected to the first housing **41** so as to be movable by the first elastic member **31**. In this case, the first elastic member **31** includes a plurality of springs (not shown). For example, the first elastic member **31** connects between the lens barrel **10** and the first housing **41** at a plurality of points of the lens barrel **10**.
(34) The second elastic member **32** is connected to the first housing **41** and the second housing **42** accommodating the first housing **41**. The second elastic member **32** fixes the first housing **41** to the second housing **42** so as to be movable. The second elastic member **32** includes a plurality of springs. In detail, the second elastic member **32** includes a plate-shaped spring.
(35) In this case, the first elastic member **31** moves the lens barrel **10** relative to the sensor unit **60** in a vertical direction (a Z-axis direction) while supporting the lens barrel **10**. To this end, the first elastic member **31** includes at least four springs.
(36) In addition, the second elastic member **32** moves the lens barrel **10** relative to the sensor unit **60** in a horizontal direction (an X-axis direction and a Y-axis direction) while supporting the lens barrel **10**. To this end, the second elastic member **32** includes at least two springs.
(37) As described above, in the camera module according to the comparative example, OIS and AF are performed as the lens barrel **10** moves in the X-axis, Y-axis, and Z-axis directions. To this end, the camera module according to the comparative example requires at least six elastic members such as springs. In addition, the camera module according to the comparative example requires two spring plates for supporting the elastic member as described above. Further, the camera module according to the comparative example requires an additional member such as an elastic wire for fixing the Z-axis of the lens barrel **10**. Therefore, the camera module according to the comparative example has a complicated spring structure for moving the lens barrel in the X-axis, Y-axis and Z-axis directions.
(38) In addition, in the camera module according to the comparative example, it is necessary to manually perform an operation of bonding the respective elastic members in order to couple the elastic member with the lens barrel **10**. Accordingly, the camera module according to the comparative example has a complicated manufacturing process and requires a long manufacturing time.
(39) In addition, the camera module according to the comparative example provides a tilt function of the lens barrel **10**, but has a structure in which tilt correction of an image is substantially difficult. That is, even though the lens barrel **10** rotates with respect to the sensor unit **60**, an image incident on the sensor unit **60** does not change, and thus the tilt correction of the image is difficult, and further, the tilt function itself is unnecessary.
(40) Hereinafter, a substrate for an image sensor, a camera module, and a camera device including the same according to an embodiment will be described.
(41) FIG. **2** is a view showing a camera device according to an embodiment.
(42) Referring to FIG. **2**, the camera device according to the embodiment includes a lens barrel **100**, a lens assembly **200**, a housing **300**, an infrared cut filter part **400**, driving parts **510**, **520**, **530**, an image sensor substrate **600**, an image sensor **700**, and a flexible circuit board **800**.
(43) The lens barrel **100** accommodates the lens assembly **200**.
(44) The lens barrel **100** may include a receiving groove for accommodating the lens assembly **200**. The receiving groove may have a shape corresponding to the lens assembly **200**.
(45) The lens barrel **100** may have a rectangular tube shape or a cylindrical shape. That is, the outer periphery of the lens barrel **100** may have a rectangular tube shape or a cylindrical shape, but the embodiment is not limited thereto.
(46) The lens barrel **100** is connected to the housing **300**. The lens barrel **100** is accommodated in the housing **300**. The lens barrel **100** may be coupled to the housing **300** by a separate coupling member (not shown).

(47) The lens barrel **100** may include an open region in an upper part thereof. Preferably, the lens barrel **100** may include a light entrance groove that is open to an object side. The light entrance groove may expose the lens assembly **200**. In addition, an image may be incident on the lens assembly **200** via the light entrance groove.
(48) The lens assembly **200** is disposed in the lens barrel **100**. The lens assembly **200** is accommodated in the accommodating groove provided in the lens barrel **100**. The lens assembly **200** may be inserted into and fixed to the accommodating groove of the lens barrel **100**. The lens assembly **200** may have a circular outer shape. For example, the lens assembly **200** may have a circular shape when viewed from a top side, but the embodiment is not limited thereto. That is, the lens assembly **200** may have a rectangular shape when viewed from the top side.
(49) The lens assembly **200** includes a plurality of lenses. For example, the lens assembly **200** may include first to fourth lenses. The first to fourth lenses may be sequentially stacked. In addition, a spacer (not shown) may be interposed between the lenses. The spacer may space a distance between the lenses constant. In the above, the lens assembly **200** has been described as including four lenses, but the embodiment is not limited thereto. For example, the lens assembly **200** may include one to three lenses, or may include five or more lenses.
(50) The housing **300** accommodates the lens barrel **100**. The housing **300** fixes a position of the lens barrel **100** via a separate fixing member (not shown). That is, according to the comparative example, the lens barrel is coupled to be movable with respect to the housing. Unlike this, in an embodiment, the housing **300** may be firmly fixed via the fixing member such that the lens barrel **100** does not move in the housing **300**. Accordingly, the position of the lens barrel **100** in the housing **300** is always fixed. Accordingly, in the embodiment, since the lens barrel **100** is always fixed at the same position, it is possible to solve a problem of distortion of an optical axis caused by warping of the lens barrel, etc., thereby improving reliability
(51) The housing **300** may be formed of plastic or metal. The housing **300** may have a rectangular tube shape.
(52) The infrared cut-off filter part **400** may be disposed at a lower end of the lens barrel **100**. The infrared cut-off filter part **400** may be fixedly disposed on a separate substrate (not shown), and accordingly, it may be coupled to the lens barrel **100**. The infrared cut-off filter part **400** may block light having an excessive long wavelength flowing into the image sensor **700**.
(53) The infrared cut-off filter **400** may be formed by alternately depositing titanium oxide and silicon oxide on an optical glass. In this case, thicknesses of the titanium oxide and the silicon oxide constituting the infrared cut-off filter **400** may be appropriately adjusted in order to block infrared rays.
(54) The driving parts **510**, **520**, and **530** move the image sensor substrate **600** relative to the fixed lens barrel **100**. The driving parts **510**, **520**, and **530** move the image sensor substrate **600** relative to the fixed housing **300**. The driving part **510**, **520**, and **530** move the image sensor substrate **600** with respect to the fixed lens assembly **200**.
(55) To this end, the driving parts **510**, **520**, and **530** may move the image sensor substrate **600** relative to magnetic force. The driving parts **510**, **520**, and **530** may include a first driving part **510**, a second driving part **520**, and a third driving part **530**.
(56) The first driving part **510** is attached to the image sensor substrate **600**. Preferably, the first driving part **510** may be attached to a lower surface of the image sensor substrate **600**. More preferably, the first driving part **510** may be attached to a lower surface of the insulating layer **610** constituting the image sensor substrate **600**. The first driving part **510** may include a magnet. For example, the first driving part **510** may include a permanent magnet. In this case, the magnet constituting the first driving part **510** may have a plate shape. Accordingly, the first driving part **510** may include an upper surface, a lower surface, and side surfaces.
(57) The second driving part **520** may be disposed on a bottom surface of the housing **300**. Preferably, the second driving part **520** may be disposed on the bottom surface of the housing **300**

overlapped with the image sensor substrate **600** in a vertical direction. The second driving part **520** may include a coil. The second driving part **520** may receive a driving signal and generate a magnetic field according to the driving signal.
(58) In this case, the first driving part **510** and the second driving part **520** may face each other. That is, the first driving part **510** and the second driving part **520** may be disposed to be overlapped with each other in the vertical direction. The first driving part **510** and the second driving part **520** may be disposed side by side in the horizontal direction. That is, the lower surface of the first driving part **510** and an upper surface of the second driving part **520** may be disposed to face each other. A separation distance between the first driving part **510** and the second driving part **520** may be 50 μm to about 1000 μm, but the embodiment is not limited thereto
(59) A magnetic force may be generated between the first driving part **510** and the second driving part **520**. Accordingly, the image sensor substrate **600** may move in a X-axis direction and a Y-axis direction by a force generated between the second driving part **520** and the first driving part **510**. Preferably, the image sensor substrate **600** may move in the X-axis direction or the Y-axis direction according to the direction of the current applied to the second driving part **520**. Also, the image sensor substrate **600** may be tilted (or rotated) according to the direction of the current applied to the second driving part **520**. To this end, the second driving part **520** and the first driving part **510** may include a plurality of magnets and a plurality of coils, respectively.
(60) That is, the first driving part **510** may include a plurality of magnets spaced apart from each other by a predetermined interval. In addition, the second driving part **520** may include a plurality of coils overlapping each of the plurality of magnets constituting the first driving part **510** in the vertical direction.
(61) According to an exemplary embodiment, the third driving part **530** may be attached to the image sensor substrate **600**. Preferably, the third driving part **530** may be attached to a sidewall of the image sensor substrate **600**. Specifically, the third driving part **530** may be attached to a side surface of a support layer **660** of the image sensor substrate **600**. In this case, at least a part of the third driving part **530** may be disposed to overlap the first driving part **510** in the horizontal direction. The third driving part **530** may be disposed perpendicular to the first driving part **510**. That is, an upper surface of the third driving part **530** may be disposed to face a side surface of the first driving part **510**. Accordingly, the image sensor substrate **600** may move in the Z-axis direction by the force between the third driving part **530** and the first driving part **510**. In this case, the third driving part **530** may be disposed in a region overlapping the first driving part **510** in the horizontal direction. That is, the third driving part **530** may include a coil wound around the support layer **660**.
(62) In addition, in another embodiment, the third driving part **530** may be arranged to overlap and align with at least a part of the first driving part **510** in a vertical direction. In this case, the third driving part **530** may be formed of an electromagnet. That is, the third driving part **530** formed of an electromagnet generates attractive or repulsive force to the first driving part **510**, so that the image sensor substrate **600** can be moved upward or downward in a Z-axis.
(63) That is, the second driving part **520** may be disposed to move the first driving part **510** in a left direction, a right direction, or rotationally. Also, the third driving part **530** may be disposed to move the first driving part **510** in an upward direction or a downward direction.
(64) An arrangement structure of the first to third driving parts and an operation thereof will be described in detail below.
(65) The image sensor substrate **600** is a substrate on which the image sensor **700** is mounted. In detail, the image sensor substrate **600** may be driven by the driving parts **510**, **520**, and **530** to move the image sensor **700** in the X, Y, and Z-axis directions. In addition, the image sensor substrate **600** may be driven by the driving parts **510**, **520**, and **530** to tilt the image sensor **700**.
(66) The image sensor substrate **600** may be disposed to be spaced apart from the bottom surface of the housing **300** by a predetermined distance. In addition, the image sensor substrate **600** may

move the mounted image sensor **700** with respect to the housing **300**.

(67) To this end, the image sensor substrate **600** may include a spring plate **610**, a first bonding sheet **620**, an insulating layer **630**, a conductive pattern portion **650**, and a support layer **660**, but is not limited thereto. That is, a second bonding sheet **640** may be further selectively formed between the insulating layer **630** and the conductive pattern portion **650**. Accordingly, the conductive pattern portion **650** may be directly formed on the insulating layer **630**, or alternatively, it may be formed on the second bonding sheet **640**. Also, when the second bonding sheet **640** is formed, the second bonding sheet **640** may have a shape corresponding to the conductive pattern portion **650**. That is, the second bonding sheet **640** may be a seed layer of the conductive pattern portion **650**. Accordingly, when the conductive pattern portion **650** is patterned, the second bonding sheet **640** may also be removed together with the conductive pattern portion **650**, and accordingly, the second bonding sheet **640** may have the same shape as the conductive pattern part **650**. That is, a planar area of the second bonding sheet **640** may be the same as a planar area of the conductive pattern portion **650**.

(68) As a result, the second bonding sheet **640** may be removed, but may be formed between the insulating layer and the conductive pattern portion in some cases. In addition, when the second bonding sheet is formed, the planar area of the second bonding sheet may have the same planar area as the planar area of the conductive pattern portion **650**. Also, in some cases, a planar area of the second bonding sheet may be the same as a planar area of the insulating layer. Hereinafter, it will be described assuming that the second bonding sheet is included.

(69) The spring plate **610** supports the insulating layer **630** and the conductive pattern portion **650** constituting the image sensor substrate **600**, and to move the image sensor **700** disposed on the image sensor substrate **600** in X-axis, Y-axis, and Z-axis directions. To this end, the spring plate **610** may include at least one elastic member. Preferably, the spring plate **610** may include a plurality of elastic members. For example, the spring plate **610** may include four elastic members.

(70) The spring plate **610** may be formed of a metal material such as stainless steel (STS) or invar, but is not limited thereto. For example, the spring plate **610** may be formed of another metal material of a spring material having an elastic force in addition to the above material. That is, the spring plate **610** may have a certain elastic force, accordingly, while elastically supporting the substrate **600** for the image sensor, the image sensor substrate **600** may be moved in the X-axis, Y-axis, and Z-axis directions.

(71) An insulating layer **630** is disposed on the spring plate **610**. In this case, the first bonding sheet **620** may be disposed between the spring plate **610** and the insulating layer **630**. The first bonding sheet **620** may be disposed between the spring plate **610** and the insulating layer **630** to provide adhesion. That is, the first bonding sheet **620** may fix the insulating layer **630** on the spring plate **610**. To this end, the first bonding sheet **620** may be formed of a double-sided adhesive film. The first bonding sheet **620** may be formed of an epoxy or acrylic adhesive.

(72) An insulating layer **630** is disposed on the first bonding sheet **620**. That is, the insulating layer **630** may be attached to the spring plate **610** by the first bonding sheet **620**.

(73) The insulating layer **630** is a substrate for forming the conductive pattern portion **650**. The insulating layer **630** may include all of a printed circuit board, a wiring board, and an insulating substrate made of an insulating material capable of forming the conductive pattern portion **650** on the surface.

(74) The insulating layers **610** may be rigid or flexible. For example, the insulating layer **610** may include glass or plastic. Specifically, the insulating layer **610** may include a chemically tempered/semi-tempered glass, such as soda lime glass, aluminosilicate glass, etc., a tempered or flexible plastic such as polyimide (PI), polyethylene terephthalate (PET), propylene glycol (PPG), polycarbonate (PC), etc., or sapphire.

(75) Further, the insulating layer **610** may include an optically isotropic film. For example, the insulating layer **610** may include cyclic olefin copolymer (COC), cyclic olefin polymer (COP),

optically isotropic PC, optically isotropic polymethylmethacrylate (PMMA), etc.
(76) In this case, the insulating layer **610** may be partially bent while having a curved surface. That is, the insulating layer **610** may partially have a plane and may partially be bent while having a curved surface. In addition, the insulating layer **610** may be a flexible substrate having flexibility. Further, the insulating layer **610** may be a curved or bended substrate.
(77) A conductive pattern portion **650** is disposed on the insulating layer **630**. The conductive pattern portion **650** may include a lead pattern part spaced apart from each other by a predetermined distance on the insulating layer **630**. For example, the conductive pattern portion **650** may include a first lead pattern part connected to the image sensor **700** and a second lead pattern part connected to the flexible circuit board **800**. In addition, the conductive pattern portion **650** may include an extension pattern part connecting the first lead pattern part and the second lead pattern part. The first lead pattern part, the second lead pattern part, and the extension pattern part will be described in detail below.
(78) The conductive pattern portion **650** is a wiring for transmitting an electrical signal, and may be formed of a metal material having high electrical conductivity. For this, the conductive pattern part **640** may be formed of at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn). In addition, the circuit pattern **112** may be formed of paste or solder paste including at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titan
(79) Preferably, the conductive pattern portion **650** serves as a wiring for transmitting an electrical signal, and may be formed of a metallic material having an elastic force movable in the X-axis, Y-axis, and Z-axis directions by interlocking with the elastic member of the siring plate **610**. To this end, the conductive pattern portion **650** may be formed of a metal material having a tensile strength of 1000 MPa or more. For example, the conductive pattern portion **650** may be a binary alloy or ternary alloy containing copper. For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-nickel (Ni). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-tin (Sn). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-beryllium (Be). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-cobalt (Co). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-nickel (Ni)-tin (Sn). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-beryllium (Be)-cobalt (Co). In addition, in addition to the metal material, the conductive pattern portion **650** may be formed of an alloy of iron (Fe), nickel (Ni), zinc (Zn), and the like having an elastic force capable of acting as a spring and having good electrical characteristics. Further, the conductive pattern portion **650** may be surface-treated with a plating layer containing a metal material such as gold (Au), silver (Ag), palladium (Pd), and the like, thereby improving electrical conductivity.
(80) On the other hand, the conductive pattern portion **650** is a typical printed circuit board manufacturing process additive process (Additive process), subtractive process (Subtractive Process), MSAP (Modified Semi Additive Process) and SAP (Semi Additive Process) method and the like, and a detailed description thereof will be omitted here.
(81) The support layer **660** may be a support substrate. A third driving part **530** may be mounted on the support layer **660**, and thus an electric signal may be applied to the third driving part **530**. The support layer **660** may support the spring plate **610**. Preferably, the support layer **660** may support the spring plate **610**, the first bonding sheet **620**, the second bonding sheet **640**, and the insulating layer **630**, the conductive pattern portion **650** and the image sensor **700** at a position floating from the bottom surface of the housing **300**. The support layer **660** may be a substrate on which the third driving part **530** according to an embodiment is disposed.
(82) The image sensor **700** is disposed on the image sensor substrate **600**. Preferably, the image sensor **700** is disposed on the conductive pattern portion **650** of the image sensor substrate **600**. The image sensor **700** may be mounted on the first lead pattern part constituting the conductive pattern

portion **650** of the image sensor substrate **600**.
(83) The image sensor **700** may be composed of a charge coupled device (CCD) or a complementary metal oxide semiconductor (CMOS) sensor, and may be a device that outputs a photographed object using a photoelectric conversion element and a charge coupled element as an electrical signal. The image sensor **700** may be electrically connected to the conductive pattern portion **650** on the image sensor substrate **600**, and receive an image provided from the lens assembly **200**. In addition, the image sensor **700** may convert the received image into an electrical signal and output the electrical signal. In this case, a signal output through the image sensor **700** may be transmitted to the flexible circuit board **800** via the conductive pattern portion **650**.
(84) One end of the flexible circuit board **800** may be connected to the image sensor substrate **600**. The flexible circuit board **800** may receive an electrical signal output from the image sensor **700**. The flexible circuit board **800** may have a connector at the other end. A main board (not shown) may be connected to the connector.
(85) That is, the flexible circuit board **800** may connect the camera module and the main board of the external device. Specifically, the flexible circuit board **800** may connect between the conductive pattern portion **650** of the image sensor substrate **600** of the camera module and the main substrate of the portable terminal.
(86) To this end, a region of the flexible circuit board **800** is disposed inside the housing **300**, and accordingly, may be connected to the conductive pattern portion **650** of the image sensor substrate **600**.
(87) Hereinafter, the image sensor substrate **600** according to an embodiment will be described in more detail.
(88) FIG. **3** is a view showing the spring plate shown in FIG.
(89) Referring to FIG. **3**, the spring plate **610** may elastically support the insulating layer **630** and the conductive pattern portion **650** constituting the image sensor substrate **600**.
(90) The spring plate **610** may move the image sensor **700** disposed on the image sensor substrate **600** in the X-axis, Y-axis, and Z-axis directions.
(91) To this end, the spring plate **610** may be formed of a material having elasticity. Specifically, the spring plate **610** may include a plurality of elastic members having elastic force. For example, the spring plate **610** may include four elastic members **614**, **615**, **616**, and **617**.
(92) The spring plate **610** may be formed of a metal material such as stainless steel (STS) or invar, but the embodiment is not limited thereto. For example, the spring plate **610** may be formed of another metal material of a spring material having elastic force in addition to the material. That is, the spring plate **660** may have a predetermined elastic force, and accordingly, the spring plate **610** may move the image sensor substrate **600** in the X-axis, Y-axis, and Z-axis directions while elastically supporting the image sensor substrate **600**.
(93) The spring plate **610** may include a first plate part **611**, a second plate part **612**, and elastic members **614**, **615**, **616**, and **617**.
(94) Specifically, the spring plate **610** may have a first plate part **611** disposed at the center thereof. In addition, the second plate part **612** may be disposed surrounding the first plate part **611** at a position spaced apart from the first plate part **611** by a predetermined distance.
(95) That is, the spring plate **610** may include the first plate part **611** and the second plate part **612**. And, an open region **613** is formed between the first plate part **611** and the second plate part **612**. Preferably, the first plate part **611** and the second plate part **612** may be separated from each other.
(96) The elastic members **614**, **615**, **616**, and **617** may have one end connected to the first plate part **611** and the other end connected to the second plate part **612**. To this end, the elastic members **614**, **615**, **616**, **617** may include a first elastic member **614**, a second elastic member **615**, a third elastic member **616**, and a fourth elastic member **617**.
(97) The first elastic member **614** may connect a first corner region of the first plate part **611** and a first corner region of the second plate part **612**. The first elastic member **614** has one end connected

to the first corner region of the first plate part **611** and the other end connected to the first corner region of the second plate part **612**. And, the first elastic member **614** may elastically connect them. The first corner region may be a corner portion located at an upper left of each plate.
(98) The second elastic member **615** may connect a second corner region of the first plate part **611** and a second corner region of the second plate part **612**. The second elastic member **615** has one end connected to the second corner region of the first plate part **611** and the other end connected to the second corner region of the second plate part **612**. And, the second elastic member **615** may elastically connect them. The second corner region may be a corner portion located at an upper right of each plate.
(99) The third elastic member **616** may connect a third corner region of the first plate part **611** and a third corner region of the second plate part **612**. The third elastic member **616** has one end connected to the third corner region of the first plate part **611** and the other end connected to the third corner region of the second plate part **612**. And, the third elastic member **616** may elastically connect them. The third corner region may be a corner portion located at the lower left of each plate.
(100) The fourth elastic member **617** may connect a fourth corner region of the first plate part **611** and a fourth corner region of the second plate part **612**. The fourth elastic member **617** has one end connected to the fourth corner region of the first plate part **611** and the other end connected to the fourth corner region of the second plate part **612**. And, the fourth elastic member **617** may elastically connect them. The fourth corner region may be a corner portion located at the lower right of each plate.
(101) Meanwhile, in the drawings, each of the first plate part **611**, the second plate part **612**, and the elastic members **614**, **615**, **616**, and **617** is illustrated as being configured separately, but is not limited thereto. That is, the first plate part **611**, the second plate part **612**, and the elastic members **614**, **615**, **616**, and **617** may be integrally formed with each other. That is, the spring plate **610** may form an open region **613** on the plate of the plate-like member in the remaining portions except for portions corresponding to the elastic members **614**, **615**, **616**, and **617**.
(102) Such the spring plate **610** may have a thickness of 10 μm to 100 μm. For example, the spring plate **610** may have a thickness of 20 μm to 70 μm. For example, the spring plate **610** may have a thickness of 40 μm to 50 μm. When the thickness of the spring plate **660** exceeds 100 μm, a thickness of the image sensor substrate **600** may increase. In addition, when the thickness of the spring plate **610** is smaller than 10 μm, a stress generated during movement of the image sensor substrate **600** may not be sufficiently maintained. Preferably, the spring plate **610** is set to have a thickness of 50 μm±10 μm to maintain a stress of 600 MPa or more.
(103) In addition, the elastic members **614**, **615**, **616**, and **617** have a length equal to or greater than a predetermined level. When the length of the elastic members **614**, **615**, **616**, and **617** is too long, there is a problem that a volume of the spring plate **610** becomes large, and when the length of the elastic members **614**, **615**, **616**, and **617** is too short, it may not be possible to stably and elastically support the image sensor substrate **600**. Therefore, the elastic members **614**, **615**, **616**, and **617** have a length of 50 μm to 100 μm. At this time, the length of the elastic members **614**, **615**, **616**, and **617** is larger than the width of the open region **613**. That is, the elastic members **614**, **615**, **616**, and **617** may be formed to have a shape of a plurality of bent springs in the open region **613**.
(104) In addition, a plurality of slits **618** are formed in the first plate part **611**. The plurality of slits **618** may be spaced apart from each other by a predetermined distance on the first plate part **61**. The plurality of slits **618** may be formed for reducing a weight of the spring plate **610**. In addition, the plurality of slits **618** may be formed for flatness of the spring plate **610**. That is, the insulating layer **630**, the conductive pattern portion **650**, and the image sensor **700** are disposed on the first plate part **611**. The first driving part is disposed below the first plate part **611**. In this case, flatness of the first driving part or the image sensor **700** directly affects the reliability of the camera module, and the image quality may be deteriorated as the flatness is worsen. Therefore, in the embodiment, not

only the weight of the spring plate **610** may be reduced but also the flatness may be maintained by forming the plurality of slits **618** in the first plate part **611**, thereby improving overall reliability of the camera module.

(105) In addition, a plurality of driving part mounting grooves (not shown) may be formed on a lower surface of the first plate part **611**. The plurality of driving part mounting grooves may be formed by removing a part of the lower surface of the first plate part **611**, for maintaining the flatness of the first driving part **510** attached to the lower surface of the first plate part **611**,

(106) FIG. **4** is a view showing the first and second bonding sheets shown in FIG. **2**.

(107) Referring to FIG. **4**, a first bonding sheet **620** is disposed on the spring plate **610**. In this case, the first bonding sheet **620** has a planar shape corresponding to the planar shape of the spring plate **610**.

(108) That is, the first bonding sheet **620** includes a first bonding part **621** disposed on the first plate part **611** of the spring plate **610** and a second bonding part **622** disposed on the second plate part **612** of the spring plate **610**. In addition, the first bonding sheet **620** may include an open region **623** between the first bonding part **621** and the second bonding part **622**.

(109) The second bonding part **622** is disposed surrounding the first bonding part **621** at a position spaced apart from the first bonding part **621** by a predetermined distance. In this case, the second bonding part **622** does not directly contact the first bonding part **621**. Accordingly, the first bonding part **621** and the second bonding part **622** may be separated from each other through the open region **623**.

(110) Meanwhile, the open region **623** of the first bonding sheet **620** may overlap the open region **613** of the spring plate **610** in a vertical direction. Preferably, a planar area of the open region **623** of the first bonding sheet **620** may be the same as a planar area of the open region **613** of the spring plate **610**. In addition, the first bonding sheet **620** does not overlap with the elastic members **614**, **615**, **616**, **617** of the spring plate **610** in the vertical direction. Accordingly, the first plate part **611** and the second plate part **612** may be covered by the first bonding sheet **620**, and the elastic members **614**, **615**, **616**, and **617** may be exposed by the open region **623** of the first bonding sheet **620**.

(111) The first bonding sheet **620** may be formed of a double-sided adhesive film. The first bonding sheet **620** may be formed of an epoxy or acrylic adhesive or a thermosetting adhesive film.

(112) The first bonding sheet **620** may have a thickness of 25 μm.

(113) FIG. **5** is a view showing an insulating layer according to an embodiment.

(114) Referring to FIG. **5**, the insulating layer **630** is disposed on the first bonding sheet **620**. The insulating layer **630** may have a planar shape corresponding to the planar shape of the first bonding sheet **620**.

(115) That is, the insulating layer **630** may include a first insulating part **631** disposed on the first bonding part **621** and a second insulating part **632** disposed on the second bonding part **622**. In addition, the insulating layer **630** may include an open region **633** between the first insulating part **631** and the second insulating part **632**.

(116) The second insulating part **632** is disposed surrounding the first insulating part **631** at a position spaced apart from the first insulating part **631** by a predetermined distance. In this case, the second insulating part **632** does not directly contact the first insulating part **631**. Accordingly, the first insulating part **631** and the second insulating part **632** may be separated from each other through the open region **633**.

(117) Meanwhile, the open region **633** of the insulating layer **630** may overlap the open region **613** of the spring plate **610** in a vertical direction. Preferably, a planar area of the open region **633** of the insulating layer **630** may be the same as the planar area of the open region **613** of the spring plate **610**. In addition, the insulating layer **630** does not overlap with the elastic members **614**, **615**, **616**, **617** of the spring plate **610** in the vertical direction. That is, the elastic members **614**, **615**, **616**, and **617** may be exposed through the open region **623** of the first bonding sheet **620** and the open

region **633** of the insulating layer **630**.
(118) The insulating layer **630** may have a thickness of 20 μm to 100 μm. For example, the insulating layer **630** may have a thickness of 25 μm to 50 μm. For example, the insulating layer **630** may have a thickness of 30 μm to 40 μm. When the thickness of the insulating layer **630** exceeds 100 μm, the overall thickness of the image sensor substrate **600** may increase. When the thickness of the insulating layer **630** is less than 20 μm, it may be difficult to dispose the image sensor **700**. When the thickness of the insulating layer **630** is less than 20 μm, it may be weak against heat/pressure, or the like in a process of mounting the image sensor, and thus it may be difficult to mount the image sensor **700** stably.
(119) Meanwhile, a second bonding sheet **640** is disposed on the insulating layer **630**.
(120) The second bonding sheet **640** has the same structure as the first bonding sheet **620**. Compared to the first bonding sheet **620**, the second bonding sheet **640** differs only in an arrangement position. Accordingly, a detailed description of the second bonding sheet **640** will be omitted. Meanwhile, although not shown in the drawing, the second bonding sheet **640** also includes an open region like the first bonding sheet **620** or the insulating layer **630**, and accordingly, the elastic members **614**, **615**, **616**, **617** is exposed.
(121) Meanwhile, in the insulating layer **630** in the first embodiment, the first insulating part and the second insulating part are separated from each other. That is, the open region **633** of the insulating layer **630** is disposed to surround the entire circumference of the first insulating part **631**.
(122) On the other hand, the open region **633** of the insulating layer **630** may be disposed partially surrounding the first insulating part **631**.
(123) FIG. **6** is a view showing a conductive pattern portion according to an embodiment.
(124) Referring to FIG. **6** the conductive pattern portion **650** may be disposed on the insulating layer **630** with a specific pattern. The conductive pattern portion **650** includes a first conductive pattern part **651** disposed on a first region of the insulating layer **630**, a second conductive pattern part **652** disposed on a second region of the insulating layer **630**, a third conductive pattern part **653** disposed on a third region of the insulating layer **630**, and a fourth conductive pattern part **654** disposed on a fourth region of the insulating layer **630**.
(125) The first conductive pattern part **651** may be disposed on a left side of an upper surface of the insulating layer **630**. That is, the first conductive pattern part **651** may be disposed on a left region of the first insulating part **631**, a left region of the second insulating part **632**, and a left region of the open region **633**.
(126) The second conductive pattern art **652** may be disposed in a right region of the upper surface of the insulating layer **630**. That is, the second conductive pattern part **652** may be disposed on a right region of the first insulating part **631**, a right region of the second insulating part **632**, and a right region of the open region **633**.
(127) The third conductive pattern part **653** may be disposed in an upper region of the upper surface of the insulating layer **630**. That is, the third conductive pattern part **653** may be disposed on an upper region of the first insulating part **631**, an upper region of the second insulating part **632**, and an upper region of the open region **633**.
(128) The fourth conductive pattern part **654** may be disposed in a lower region of the upper surface of the insulating layer **630**. That is, the fourth conductive pattern part **654** may be disposed on a lower region of the first insulating part **631**, a lower region of the second insulating part **632**, and a lower region of the open region **633**.
(129) As described above, the conductive pattern portion **650** are disposed on different regions, and accordingly, the elastic support force in the movement of the image sensor substrate **600** may be increased. That is, when the conductive pattern portion **650** is intensively disposed only in a specific region, reliability in moving the image sensor substrate **600** in a specific direction may be reduced. For example, when the conductive pattern portion **650** includes only the first and second conductive pattern parts, there is no problem with the movement of the image sensor substrate **600**

in the X-axis direction, stability may be deteriorated when the image sensor substrate **600** moves in the Y-axis direction. In addition, in this case, the conductive pattern portion **650** may be disconnected according to the periodic movement of the image sensor substrate **600**. Accordingly, in the embodiment, the conductive pattern portion **650** is distributed in each of the four regions as described above, so that the image sensor substrate **600** can be stably moved in the X-axis, Y-axis, and Z-axis.

(130) Meanwhile, the conductive pattern portion **650** may include a first lead pattern part **655** connected to the image sensor **700** and a second lead pattern part **656** connected to the flexible circuit board **800**. In addition, the conductive pattern portion **650** may include an extension pattern part **657** connecting the first lead pattern part **655** and the second lead pattern part **656** to each other.

(131) The first lead pattern part **655** is disposed on the first insulating part **631** of the insulating layer **630**. The first lead pattern part **655** may be disposed in an outer region of the first insulating part **631**. That is, the image sensor mounting region on which the image sensor **700** is mounted may be formed in the first insulating part **631**. In this case, the image sensor mounting region may be a central region of the first insulating part **631**. Accordingly, the first lead pattern part **655** may be disposed around the image sensor mounting region of the first insulating part **631**.

(132) The second lead pattern part **656** is disposed on the second insulating part **632** of the insulating layer **630**. The second lead pattern parts **656** may be respectively disposed on the second insulating part **632**. In this case, the first lead pattern part **655** and the second lead pattern part **656** may be disposed to face each other on the first insulating part **631** and the second insulating part **632**. That is, the number of first lead pattern parts **655** may be the same as the number of second lead pattern parts **656**. In addition, the first lead pattern parts **655** may be disposed to face the second lead pattern part **656**, respectively.

(133) Meanwhile, the extension pattern part **657** may be disposed between the first lead pattern part **655** and the second lead pattern part **656**.

(134) The extension pattern part **657** may have one end connected to the first lead pattern part **655** and the other end connected to the second lead pattern part **656** facing the first lead pattern part **655**.

(135) In this case, the extension pattern part **657** may be disposed on the open region of the insulating layer **630**. Accordingly, the extension pattern part **657** may be disposed to float in the open region of the insulating layer, the open region of the first bonding sheet, the open region of the second bonding sheet, and the open region of the spring plate **610**. Here, the meaning that the extension pattern part **657** is floating means that there is no support part supporting the extension pattern part **657** at the bottom. Accordingly, the extension pattern part **657** may be disposed in a state floating in the air.

(136) A length of the extension pattern part **657** is greater than a linear distance between the first lead pattern part **655** and the second lead pattern part **656**. That is, the extension pattern part **657** may be formed to have a structure that is bent a plurality of times between the first lead pattern part **655** and the second lead pattern part **656**. Preferably, the extension pattern part **657** may be formed in a spring shape between the first lead pattern part **655** and the second lead pattern part **656**.

(137) At this time, the extension pattern part **657** may be formed by etching to have the above shape through an additive process, a subtractive process, a modified semi-additive process (MSAP), a semi-additive process (SAP), and the like. Preferably, the extension pattern part **657** may be formed simultaneously with the first lead pattern part **655** and the second lead pattern part **656**. More preferably, the extension pattern part **657** may be integrally formed with the first lead pattern part **655** and the second lead pattern part **656**.

(138) Meanwhile, the thickness of the conductive pattern portion **650** including the extension pattern part **657**, the first lead pattern part **655**, and the second lead pattern part **656** may be 10 μm to 50 μm. For example, the thickness of the conductive pattern portion **650** may be 30 μm to 40

μm. In this case, when the thickness of the conductive pattern portion **650** is less than 10 μm, the conductive pattern portion **650** may be broken when the image sensor substrate **600** moves. In addition, when the thickness of the conductive pattern portion **650** is greater than 50 μm, elastic force of the extension pattern part **657** may be lowered, thereby hindering the mobility of the image sensor substrate **600**. Accordingly, in the embodiment, the thickness of the conductive pattern portion **650** is set to be 35 μm±5 μm such that the image sensor substrate **600** may be stably moved.

(139) In addition, the length of the extension pattern part **657** is set to have at least 1.5 times the linear distance between the first lead pattern part **655** and the second lead pattern part **656**. In addition, the length of the extension pattern part **657** is set to be 20 times or less of the linear distance between the first lead pattern part **655** and the second lead pattern part **656**. Preferably, the length of the extension pattern part **657** is set to be less than 4 times the linear distance between the first lead pattern part **655** and the second lead pattern part **656**.

(140) The linear distance between the first lead pattern part **655** and the second lead pattern part **656** may be 1.5 mm.

(141) In this case, when the length of the extension pattern part **657** is smaller than 1.5 times the linear distance between the first lead pattern part **655** and the second lead pattern part, the mobility of the image sensor substrate **600** may be reduced due to the decrease in the elastic force of the extension pattern part **657**. In addition, when the length of the extension pattern part **657** is greater than 20 times the linear distance, a resistance increases as a signal transmission distance is increased by the extension pattern part **657**, and accordingly, noise may be included in a signal transmitted via the extension pattern part **657**. Accordingly, in order to minimize noise generation, the length of the extension pattern part **657** is set to be 4 times or less the linear distance between the first lead pattern part **655** and the second lead pattern part **656**.

(142) In addition, the length of the extension pattern part **657** is smaller than the lengths of the elastic members **614**, **615**, **616**, and **617**.

(143) Meanwhile, the conductive pattern portion **650** as described above is a wiring for transmitting an electrical signal, and may be formed of a metal material having high electrical conductivity. For this, the conductive pattern portion **650** may be formed of at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn). In addition, the conductive pattern portion **650** may be formed of paste or solder paste including at least one metal material selected from among gold (Au), silver (Ag), platinum (Pt), titanium (Ti), tin (Sn), copper (Cu), and zinc (Zn), which are excellent in bonding strength.

(144) Preferably, the conductive pattern portion **650** serves as a wiring for transmitting an electrical signal, and may be formed of a metallic material having an elastic force movable in the X-axis, Y-axis, and Z-axis directions by interlocking with the elastic member. To this end, the conductive pattern portion **650** may be formed of a metal material having a tensile strength of 1000 MPa or more. For example, the conductive pattern portion **650** may be a binary alloy or ternary alloy containing copper. For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-nickel (Ni). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-tin (Sn). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-beryllium (Be). For example, the conductive pattern portion **650** may be a binary alloy of copper (Cu)-cobalt (Co). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-nickel (Ni)-tin (Sn). For example, the conductive pattern portion **650** may be a ternary alloy of copper (Cu)-beryllium (Be)-cobalt (Co). In addition, in addition to the metal material, the conductive pattern portion **650** may be formed of an alloy of iron (Fe), nickel (Ni), zinc (Zn), and the like having an elastic force capable of acting as a spring and having good electrical characteristics. Further, the conductive pattern portion **650** may be surface-treated with a plating layer containing a metal material such as gold (Au), silver (Ag), palladium (Pd), and the like,

thereby improving electrical conductivity.
(145) FIG. **7** is a plan view of an image sensor substrate according to an embodiment.
(146) Referring to FIG. **7**, the first bonding sheet **620** is disposed on the spring plate **610**. In addition, the insulating layer **630** is disposed on the first bonding sheet **620**. In addition, the second bonding sheet **630** is disposed on the insulating layer **630**. In addition, the conductive pattern portion **650** is disposed on the second bonding sheet **630**. In this case, the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet **630** each include an open region, and each of the open regions may overlap each other in a vertical direction. Therefore, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** disposed at a lowermost part may be exposed through the open regions of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet.
(147) The extension pattern part **657** of the conductive pattern portion **650** is also disposed on the open region of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet **630**.
(148) That is, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** and the extension pattern part **657** of the conductive pattern portion **650** are disposed in the open regions of the spring plate **610**, the first bonding sheet **620**, and the insulating layer **630**, and the second bonding sheet **630**.
(149) In this case, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** and the extension pattern part **657** of the conductive pattern portion **650** do not overlap each other in the vertical direction. For example, the elastic members **614**, **615**, **616**, **617** of the spring plate **610** are disposed in a first region of the open regions of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, the second bonding sheet **630**. And, the extension pattern part **657** of the conductive pattern portion **650** is disposed in a second region of the open regions of the spring plate **610**, the first bonding sheet **620**, the insulating layer **630**, and the second bonding sheet **630**. The first region of the open regions may be a corner region of the open regions, and the second region of the open regions may be a remaining region except for the corner region.
(150) That is, the image sensor substrate **600** not only moves in the X-axis direction and the Y-axis direction, but also moves in the Z-axis direction. At this time, there is a difference between the elastic modulus of the elastic members **614**, **615**, **616**, and **617** and the elastic modulus of the extension pattern part **657**. Accordingly, when the image sensor substrate **600** moves in the Z-axis direction, the movement distances of the elastic members **614**, **615**, **616**, **617** and the extension pattern part **657** appear different from each other. Accordingly, when the image sensor substrate **600** moves in the Z-axis direction, the elastic members **614**, **615**, **616**, **617** and the extension pattern part **657** may contact each other. Accordingly, a problem (for example, a short) may occur in electrical reliability. Accordingly, in the embodiment, as described above, the elastic members **614**, **615**, **616**, **617** and the extension pattern part **657** are not aligned with each other in the vertical direction. Thereby, the electrical reliability problem can be solved.
(151) FIG. **8** is a view showing an arrangement structure of a first driving part according to an exemplary embodiment.
(152) Referring to FIG. **8**, the first driving part **510** may include a plurality of magnets. The first driving part **510** may include a first-first driving part **511**, a first-second driving part **512**, a first-third driving part **513**, and a first-fourth driving part **514**.
(153) The first-first driving part **511** may be disposed in one region of the first plate part **611** of the spring plate **610**. Preferably, the first-first driving part **511** may be disposed in a first corner region of the first plate part **611**. For example, if the lower surface of the first plate part **611** is divided into four planes, the first-first driving part **511** may be disposed in a second quadrant. In this case, the first-first driving part **511** may be arranged such that the N pole and the S pole are arranged in a horizontal direction. For example, the first-first driving part **511** may be disposed under the lower surface of the first plate part **611** such that the N pole and the S pole are positioned in the X-axis

direction.

(154) The first-second driving part **512** may be disposed in one region of the first plate part **611** of the spring plate **610**. Preferably, the first-second driving part **512** may be disposed in the second corner region of the first plate part **611**. For example, if the lower surface of the first plate part **611** is divided into four planes, the first-second driving part **512** may be disposed in a first quadrant. In this case, the first-second driving part **512** may be arranged such that the N pole and the S pole are arranged in a vertical direction. For example, the first-second driving part **512** may be disposed on the lower surface of the first plate part **611** such that the N pole and the S pole are positioned in the Y-axis direction. In this case, the first-first driving part **511** and the first-second driving part **512** may be disposed such that the N pole and the S pole are perpendicular to each other. For example, the first-second driving part **512** may be disposed while being rotated 90 degrees to the right with respect to the first-first driving part **511**.

(155) The first-third driving part **513** may be disposed in one region of the first plate part **611** of the spring plate **610**. Preferably, the first-third driving parts **513** may be disposed in the third corner region of the first plate part **611**. For example, when the lower surface of the first plate part **611** is divided into four planes, the first-third driving parts **513** may be disposed in a fourth quadrant. In this case, the first-third driving part **513** may be arranged such that the N pole and the S pole are arranged in a horizontal direction. For example, the first-third driving part **513** may be disposed on the lower surface of the first plate part **611** such that the N pole and the S pole are positioned in the X-axis direction. In this case, the first-second driving part **512** and the first-third driving part **513** may be disposed such that the N pole and the S pole are perpendicular to each other. Also, the first-first driving part **511** and the first-third driving part **513** may be disposed such that the N pole and the S pole are in opposite directions. For example, the first-third driving part **513** may be disposed in a state rotated 90 degrees in a right direction with respect to the first-second driving part **512**. The first-third driving part **513** may be disposed in a state rotated 180 degrees in the right direction with respect to the first-first driving part **511**.

(156) The first-fourth driving part **514** may be disposed in one region of the first plate part **611** of the spring plate **610**. Preferably, the first-fourth driving part **514** may be disposed in the fourth corner region of the first plate part **611**. For example, when the lower surface of the first plate part **611** is divided into four planes, the first-fourth driving part **514** may be disposed in the third quadrant. In this case, the first-fourth driving pars **514** may be arranged such that the N pole and the S pole are arranged in a vertical direction. For example, the first-fourth driving part **514** may be disposed on the lower surface of the first plate part **611** such that the N pole and the S pole are positioned in the Y-axis direction. In this case, the first-third driving part driving part **513** and the first-fourth driving part **514** may be disposed such that the N pole and the S pole are perpendicular to each other. Also, the first-second driving part **512** and the first-fourth driving part **514** may be disposed such that the N pole and the S pole are in opposite directions to each other. For example, the first-fourth driving part **514** may be disposed in a state rotated 90 degrees to the right direction with respect to the first-second driving part **513**. The first-fourth driving part **514** may be disposed in a state rotated 180 degrees in the right direction with respect to the first-second driving part **512**. The first-fourth driving part **514** may be disposed in a state rotated by 90 degrees to the left direction or 270 degrees to the right direction with respect to the first-first driving part **511**.

(157) Here, the first-first driving part **511** and the first-third driving part **513** in which the N pole and S pole arranged in the X-axis direction are a first driving part of a first group for moving the image sensor substrate **600** in the X-axis direction. In addition, the first-second driving part **512** and the first-fourth driving part **514** in which the N pole and the S pole are arranged in the Y-axis direction are a first driving part of a second group for moving the image sensor substrate **600** in the Y-axis direction. Also, any one or both of the first driving part of the first group and the first driving part of the second group may tilt the image sensor substrate **600**. This will be described in more detail below.

(158) As described above, in the embodiment, the first driving part **510** may be disposed such that the same poles are adjacent to each other with respect to the center C so as to receive a force from the third driving part **530**. In addition, the driving parts adjacent to each other among the first driving parts **510** make the arrangement directions of the N pole and the S pole perpendicular to each other.

(159) Accordingly, the first driving part **510** may be arranged such that the same pole is adjacent to the center C, and the same pole surrounds an outer part. That is, an outer part of the first plate part **611** may include a left part, a right part, an upper part, and a lower part. In addition, the left part may be disposed to face the N pole of the first-first driving part **511**. In addition, the upper part may be disposed to face the N pole of the first-second driving part **512**. In addition, the right part may be disposed to face the N pole of the first-third driving part **513**. Also, the lower part may be disposed to face the N pole of the first-fourth driving part **514**.

(160) FIG. **9** is a view showing an arrangement structure of a second driving part according to an embodiment.

(161) Referring to FIG. **9**, the second driving part **520** may include a second-first driving part **521**, a second-second driving part **522**, a second-third driving part **523**, and a second-fourth driving part **524**.

(162) The second-first driving part **521** may be disposed to overlap with the first-first driving part **511** constituting the first driving part **510** in the vertical direction. In addition, the second-first driving part **521** may be disposed to correspond to the arrangement direction of the N pole and the S pole of the first-first driving part **511**. A driving direction (or movement direction) of the first-first driving part **511** is determined according to a direction of a current applied to the second-first driving part **521**. For example, when the direction of the current applied to the second-first driving part **521** is a first direction, the first-first driving part **511** may move in the −X-axis direction. Also, when the direction of the current applied to the second-first driving part **521** is a second direction opposite to the first direction, the first-first driving part **511** may move in the +X-axis direction.

(163) The second-second driving part **522** may be disposed to overlap with the first-second driving part **512** constituting the first driving part **510** in the vertical direction. In addition, the second-second driving part **522** may be disposed to correspond to the arrangement direction of the N pole and the S pole of the first-second driving part **512**. A driving direction (or movement direction) of the first-second driving part **512** is determined according to a direction of a current applied to the second-second driving part **522**. For example, when the direction of the current applied to the second-second driving part **522** is a first direction, the first-second driving part **512** may move in the −Y-axis direction. Also, when the direction of the current applied to the second-second driving part **522** is a second direction opposite to the first direction, the first-second driving part **512** may move in the +Y-axis direction.

(164) The second-third driving part **523** may be disposed to overlap with the first-third driving part **513** constituting the first driving part **510** in the vertical direction. In addition, the second-third driving part **523** may be disposed to correspond to the arrangement direction of the N pole and the S pole of the first-third driving part **513**. A driving direction (or movement direction) of the first-third driving part **513** is determined according to a direction of a current applied to the second-third driving part **523**. For example, when the direction of the current applied to the second-third driving part **523** is a first direction, the first-third driving part **513** may move in the +X-axis direction. Also, when the direction of the current applied to the second-third driving part **523** is a second direction opposite to the first direction, the first-third driving part **513** may move in the −X-axis direction.

(165) The second-fourth driving part **524** may be disposed to overlap with the first-fourth driving part **514** constituting the first driving part **510** in the vertical direction. In addition, the second-fourth driving part **524** may be disposed to correspond to the arrangement direction of the N pole and the S pole of the first-fourth driving part **514**. A driving direction (or movement direction) of the first-fourth driving part **514** is determined according to a direction of a current applied to the

second-fourth driving part **524**. For example, when the direction of the current applied to the second-fourth driving part **524** is a first direction, the first-fourth driving part **514** may move in the +Y-axis direction. Also, when the direction of the current applied to the second-fourth driving part **524** is a second direction opposite to the first direction, the first-fourth driving part **514** may move in the −Y-axis direction.

(166) Here, the second-first driving part **521** and the second-third driving part **523** are a second driving part of a first group for moving the image sensor substrate **600** in the +X-axis direction or the −X-axis direction. In addition, the second-second driving part **522** and the second-fourth driving part **524** are a second driving part of a second group for moving the image sensor substrate **600** in the +Y axis or the −Y axis direction. In addition, any one or both of the second driving part of the first group and the second driving part of the second group may tilt the image sensor substrate **600**.

(167) Hereinafter, operations of the first driving part **510** and the second driving part **520** for moving the image sensor substrate **600** in the +X-axis direction, the −X-axis direction, the +Y-axis direction, and the −Y-axis direction to be explained.

(1) Movement in the +X-Axis Direction

(168) In order to move the image sensor substrate **600** in the +X-axis direction, a current may be applied to the second-first driving part **521** and the second-third driving part **523**. To this end, a current in the first direction may be applied to the second-first driving part **521**, and a current in the second direction may be applied to the second-third driving part **523**. Accordingly, a force from the N pole to the S pole is applied to the first-first driving part **511** of the first driving part **510**, and a force from the S pole to the N pole direction is applied to the first-third driving part **513**. In addition, the image sensor substrate **600** may move in the +X-axis direction according to the driving directions of the first-first driving part **511** and the first-third driving part **513**.

(2) Movement in the −X-Axis Direction

(169) In order to move the image sensor substrate **600** in the −X-axis direction, a current may be applied to the second-first driving part **521** and the second-third driving part **523**. To this end, a current in the second direction may be applied to the second-first driving part **521**, and a current in the first direction may be applied to the second-third driving part **523**. Accordingly, a force from the S pole to the N pole is applied to the first-first driving part **511** of the first driving part **510**, and a force from the N pole to the S pole is applied to the first-third driving part **513**. In addition, the image sensor substrate **600** may move in the −X-axis direction according to the driving directions of the first-first driving part **511** and the first-third driving part **513**.

(3) Movement in the +Y-Axis Direction

(170) In order to move the image sensor substrate **600** in the +Y-axis direction, a current may be applied to the second-second driving part **522** and the second-fourth driving part **524**. To this end, a current in the first direction may be applied to the second-second driving part **522** and a current in the second direction may be applied to the second-fourth driving part **524**. Accordingly, a force from the N pole to the S pole is applied to the first-second driving part **512** of the first driving part **510**, and a force from the S pole to the N pole direction is applied to the first-fourth driving part **514**. Also, the image sensor substrate **600** may move in the +Y-axis direction according to the driving directions of the first-second driving part **512** and the first-fourth driving part **513**.

(4) Movement in the −Y Axis Direction

(171) In order to move the image sensor substrate **600** in the −Y-axis direction, a current may be applied to the second-second driving part **522** and the second-fourth driving part **524**. To this end, a current in the second direction may be applied to the second-second driving part **522** and a current in the first direction may be applied to the second-fourth driving part **524**. Accordingly, ae force from the S pole to the N pole is applied to the first-second driving part **512** of the first driving part **510**, and a force from the N pole to the S pole direction is applied to the first-fourth driving part **514**. In addition, the image sensor substrate **600** may move in the −Y-axis direction according to

the driving directions of the first-second driving part **512** and the first-fourth driving part **513**.
(5) Tilt Clockwise
(172) The tilt of the image sensor substrate **600** may include a clockwise tilt and a counterclockwise tilt. The clockwise tilt includes a first method of applying a current to the second-first driving part **521** and the second-third driving part **523**, a second method of applying a current to all of the second-second driving part **522** and the second-fourth driving part **524**, and a third method of applying a current to the second-first driving part **521**, the second-second driving part **522**, the second-third driving part **523**, and the second-fourth driving part **524**.
(173) When the first method is described, current may be applied to the second-first driving part **521** and the second-third driving part **523** in the same first direction. At this time, the N-pole and S-pole directions of the first-first driving part **511** and the first-third driving part **513** are opposite to each other. Accordingly, as the current is applied in the first direction as described above, the first-first driving part **511** may move in the +X axis, and the first-third driving part **513** may move in the −X axis, and thus Accordingly, the image sensor substrate **600** may be tilted clockwise.
(174) When the second method is described, current may be applied to the second-second driving part **522** and the second-fourth driving part **524** in the same first direction. In this case, the N-pole and S-pole directions of the first-second driving part **512** and the first-fourth driving part **514** are opposite to each other. Accordingly, as the current is applied in the first direction as described above, the first-second driving part **512** may move in the +Y axis, and the first-fourth driving part **514** may move in the −Y axis, and thus Accordingly, the image sensor substrate **600** may be tilted clockwise.
(175) When the third method is described, the current may be applied to the second-first driving part **521** and the second-third driving part **523** in the same first direction. Also, the current may be applied to the second-second driving part **522** and the second-fourth driving part **524** in the same first direction. Accordingly, the first-first driving part **511** may move in the +X axis, the first-third driving part **513** may move in the −X axis, and the first-second driving part **512** may move in the +Y axis. and the first-fourth driving part **514** may move along the −Y axis, and accordingly, the image sensor substrate **600** may be tilted clockwise.
(6) Tilt Counterclockwise
(176) The tilt of the image sensor substrate **600** may include a clockwise tilt and a counterclockwise tilt. The clockwise tilt includes a first method of applying a current to the second-first driving part **521** and the second-third driving part **523**, a second method of applying a current to the second-second driving part **522** and the second-fourth driving part **524**, and a third method of applying a current to all of the second-first driving part **521**, the second-second driving part **522**, the second-third driving part **523**, and the second-fourth driving part **524**.
(177) When the first method is described, the current may be applied to the second-first driving part **521** and the second-third driving part **523** in the same second direction. At this time, the N-pole and S-pole directions of the first-first driving part **511** and the first-third driving part **513** are opposite to each other. Accordingly, as the current is applied in the second direction as described above, the first-first driving part **511** may move in the −X axis, and the first-third driving part **513** may move in the +X axis, and thus Accordingly, the image sensor substrate **600** may be tilted counterclockwise.
(178) In the second method, current may be applied to the second-second driving part **522** and the second-fourth driving part **524** in the same second direction. In this case, the N-pole and S-pole directions of the first-second driving part **512** and the first-fourth driving part **514** are opposite to each other. Accordingly, as the current is applied in the second direction as described above, the first-second driving part **512** may move in the −Y axis, and the first-fourth driving part **514** may move in the +Y axis, and thus Accordingly, the image sensor substrate **600** may be tilted counterclockwise.
(179) When the third method is described, current may be applied to the second-first driving part

521 and the second-third driving part **523** in the same second direction. Also, a current may be applied to the second-second driving part **522** and the second-fourth driving part **524** in the same second direction. Accordingly, the first-first driving part **511** may move in the −X axis, the first-third driving part **513** may move in the +X axis, and the first-second driving part **512** may move in the −Y axis. and the first-fourth driving part **514** may move in the +Y axis, and accordingly, the image sensor substrate **600** may be tilted counterclockwise.
(180) FIG. **10** is a view showing an arrangement structure of a third driving part according to an embodiment.
(181) Referring to FIG. **10**, the third driving part **530** may be configured as a coil. The third driving part **530** may be disposed on the support layer **660** while surrounding the first driving part **510**. In this case, the third driving part **530** may be disposed to face the N poles of each of the first-first to first-fourth driving parts constituting the first driving part **510**. Accordingly, when a current in the first direction is applied to the third driving part **530**, a force in the +Z axis direction may be applied to the first driving part **510** to move in the +Z axis direction. In addition, when a current in the second direction is applied to the third driving part **530**, a force in the −Z axis direction may be applied to the first driving part **510** to move in the −Z axis direction.
(182) FIG. **11** is a view showing an arrangement structure of a third driving part according to another exemplary embodiment.
(183) Referring to FIG. **11**, the third driving part may be formed of an electromagnet. To this end, the third driving part **530** includes a magnetic body **531** and a coil part **532** surrounding the magnetic body **531**. At this time, when a current is applied to the coil part **532** in the first direction, magnetic force of N pole and S pole is generated in the magnetic body **531** based on an upper part and a lower part. In this case, the third driving part **530** is disposed under the first driving part **510** together with the second driving part **520**. In this case, the third driving part **530** is disposed to overlap at least a part of the first driving part **510** based on a center of the first driving part **510**.
(184) In other words, the third driving part **530** may be include a region overlapping a part of the S pole of the first-first driving part **511**, a region overlapping a part of the S pole of the first-second driving part **512**, a region overlapping a part of the S pole of the first-third driving part **513**, and a region overlapping a part of the S pole of the first-fourth driving part **514**. And, when a current in the first direction is applied to the third driving part **530**, an upper part of the third driving part **530** becomes the N pole and a lower part of the third driving part **530** becomes the S pole. Accordingly, an attractive force is generated between the first driving part **510** and the third driving part **530**, and the first driving part **510** may be moved in the −Z axis direction. Conversely, when a current in the second direction is applied to the third driving part **530**, the upper part of the third driving part **530** becomes the S pole and the lower part of the third driving part **530** becomes the N pole. Accordingly, a repulsive force is generated between the first driving part **510** and the third driving part **530**, and the first driving part **510** may move in the +Z-axis direction.
(185) FIG. **12** is a view showing an arrangement structure of a third driving part according to another exemplary embodiment.
(186) In FIG. **11**, the third driving part **530** is configured using only one electromagnet. In this case, a strength of the magnetic field transmitted to the first driving part **510** by one third driving part **530** may be weak. Accordingly, the third driving part **530** according to another embodiment may include a plurality of electromagnets.
(187) To this end, the third driving part **530** includes a third-first driving part **530**A, a third-second driving part **530**B, a third-third driving part **530**C, and a third-fourth driving part **530**D.
(188) The third-first driving part **530**A may include a region overlapping a part of the S pole of the first-first driving part **511** and a region overlapping a part of the S pole of the first-second driving part **512**. Preferably, the third-first driving part **530**A may be disposed to overlap a region between the first-first driving part **511** and the first-second driving part **512** in the vertical direction.
(189) The third-second driving part **530**B may include a region overlapping a part of the S pole of

the first-second driving part **512** and a region overlapping a part of the S pole of the first-third driving part **513**. Preferably, the third-second driving part **530**B may be disposed to overlap a region between the first-second driving part **512** and the first-third driving part **513** in the vertical direction.
(190) The third-third driving part **530**C may include a region overlapping a part of the S pole of the first-third driving part **513** and a region overlapping a part of the S pole of the first-fourth driving part **514**. Preferably, the third-third driving part **530**C may be disposed to overlap a region between the first-third driving part **513** and the first-fourth driving part **514** in the vertical direction.
(191) The third-fourth driving part **530**D may include a region overlapping a part of the S pole of the first-first driving part **511** and a region overlapping a part of the S pole of the first-fourth driving part **514**. Preferably, the third-fourth driving part **530**D may be disposed to overlap a region between the first-first driving part **511** and the first-fourth driving part **514** in the vertical direction.
(192) According to an embodiment, in order to implement the OIS and AF functions of the camera module, instead of moving the conventional lens barrel, the image sensor is moved with respect to the lens barrel in the X-axis, Y-axis and Z-axis directions. Accordingly, the camera module according to the embodiment may remove a complex spring structure for implementing the OIS and AF functions, and thus the structure may be simplified. In addition, by moving the image sensor according to the embodiment with respect to the lens barrel, it is possible to provide a stable structure compared to the conventional structure.
(193) In addition, according to an embodiment, the extension pattern part electrically connected to the image sensor has a spring structure and is disposed to float on a spring plate. Accordingly, the camera module may move the image sensor while elastically supporting the image sensor more stably. In addition, the length of the extension pattern part is at least 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part. Accordingly, it is possible to minimize noise generation while improving the mobility of the image sensor substrate.
(194) In addition, according to an embodiment, by preventing the elastic member and the extension pattern part from being aligned with each other in the vertical direction, it is possible to solve an electrical reliability problem that may occur due to a contact between the elastic member and the extension pattern part.
(195) In addition, in the insulating layer in the embodiment, an extension insulating part having a spring shape is disposed in a region that vertically overlaps with the extension pattern part, so that the spring plate can be removed. Accordingly, the camera module may move the image sensor with respect to the lens barrel while elastically supporting the image sensor more stably, and may minimize the product volume.
(196) In addition, in the embodiment, the width of the extension insulating part is greater than the width of the extension pattern part, so that the extension pattern part can be stably supported by the extension insulating part, and thus operation reliability can be improved.
(197) In addition, in the embodiment, the first driving part composed of a permanent magnet is disposed so that the same poles are adjacent to each other with respect to the center, while the N poles and the S poles of the permanent magnets adjacent to each other are perpendicular to each other. According to this, in the embodiment, it is possible to secure the arrangement freedom of the second driving part and the third driving part for adjusting the electromagnetic force in response to the permanent magnet, it is possible to move in the X-axis and Y-axis, and stable tilt control is also possible.
(198) FIG. **13** is a view showing a connection structure between a flexible circuit board and a board for an image sensor according to an embodiment.
(199) Referring to FIG. **13**, the flexible circuit board **800** electrically connects the image sensor substrate **600** and an external main substrate (not shown) to each other.
(200) One end of the flexible circuit board **800** may be connected to the image sensor substrate **600**. The flexible circuit board **800** may receive an electrical signal output from the image sensor

700. The flexible circuit board **800** may have a connector **810** at the other end. A main board (not shown) may be connected to the connector **810**.
(201) That is, the flexible circuit board **800** may connect the camera module and the main board of the external device. Specifically, the flexible circuit board **800** may connect between the conductive pattern portion **650** of the image sensor substrate **600** of the camera module and the main board of a portable terminal.
(202) To this end, an region of the flexible circuit board **800** is disposed inside the housing **300**, and accordingly, may be connected to the conductive pattern portion **650** of the image sensor substrate **600**.
(203) That is, the flexible circuit board **800** may include a first connector part **801**, a second connector part **803**, and a connection part **802**.
(204) The first connector part **801** may be disposed inside the housing **300**. The first connector part **801** may include a plurality of pads **804**, **805**, **806**, and **807** connected to the conductive pattern portion **650**.
(205) The first connector part **801** may be electrically connected to the second lead pattern part **656** of the conductive pattern portion **650**. That is, the plurality of pads **804**, **805**, **806**, and **807** of the first connector part **801** may be electrically connected to the second lead pattern part **656**.
(206) To this end, the first connector part **801** may include a first pad part **804** connected to the second lead pattern part **656** of the first conductive pattern part **651**. In addition, the first connector part **801** may include a second pad part **805** connected to the second lead pattern part **656** of the second conductive pattern part **652**. In addition, the first connector part **801** may include a third pad part **806** connected to the second lead pattern part **656** of the third conductive pattern part **653**. In addition, the first connector part **801** may include a fourth pad part **807** connected to the second lead pattern part **656** of the fourth conductive pattern part **654**.
(207) In this case, the first connector part **801** has a shape corresponding to the second insulating part **632** of the insulating layer **630**. Accordingly, the first connector part **801** may be disposed surrounding an upper region of the second lead pattern part **656** of the conductive pattern portion **650**, and the plurality of pads **804**, **805**, **806**, and **807** may be disposed on a lower surface of the first connector part **801**.
(208) The connection part **802** connects the first connector part **801** and the second connector part **803** to each other. A part of the connection part **802** may be disposed in the housing **300**, and may be extended therefrom to be exposed to the outside of the housing **300**.
(209) The second connector part **803** may include a connector **810** connected to the main board of the terminal.
(210) Meanwhile, the image sensor **1700** may be attached on the first insulating part **1631** of the insulating layer **1610**.
(211) In this case, an electrode **710** of the image sensor **700** faces upward and may be attached to the first insulating part **631**. In addition, a connection member **720** such as a metal wire is formed between the electrode **710** of the image sensor **700** and the first lead pattern part **655**. The connection member **720** may electrically connect the electrode of the image sensor and the first lead pattern part.
(212) FIG. **14** is a view for explaining in detail a layer structure of a conductive pattern portion according to an embodiment.
(213) Referring to FIG. **14**, the conductive pattern portion **650** is disposed on the second bonding sheet **640** disposed on the insulating layer **630**. In this case, the conductive pattern portion **650** includes the first lead pattern part **655** disposed on the first insulating part **631** of the insulating layer **630**, the second lead pattern part **656** on the second insulating part **632**, and the extension pattern part **657** connecting therebetween.
(214) In this case, each of the first lead pattern part **655**, the second lead pattern part **656**, and the extension pattern part **657** may include a metal layer **650**A and a plating layer **650**B.

(215) The metal layer **650**A may be disposed on the second bonding sheet **640**. That is, the metal layer **650**A is disposed on the second bonding sheet **640** to configure the first lead pattern part **655**, the second lead pattern part **656**, and the extension pattern part **657**, respectively.
(216) The plating layer **650**B may be disposed on the metal layer **650**A. Preferably, the plating layer **650**B may be a surface treatment layer disposed on the metal layer **650**A.
(217) The plating layer **650**B includes any one of Ni/Au alloy, gold (Au), electroless nickel immersion gold (ENIG), Ni/Pd alloy, and organic compound plating (Organic Solderability Preservative, OSP).
(218) In this case, the plating layers **650**B constituting the first lead pattern part **655** and the second lead pattern part **656** may correspond to each other. Alternatively, the plating layer **650**B constituting the extension pattern part **657** may have a thickness different from that of the plated layer **650**B constituting the first lead pattern part **655** and the second lead pattern part **656**.
(219) That is, the plating layer **650**B of the first lead pattern part **655** and the second lead pattern part **656** may be selectively formed only on upper and side surfaces of the corresponding metal layer **650**A. In contrast, the metal layer **650**A constituting the extension pattern part **657** is disposed in a floating state on the above-described open region. Accordingly, the plating layer **650**B of the extension pattern part **657** may be formed to surround the entire surface of the corresponding metal layer **650**A. That is, the plating layer **650**B of the extension pattern part **657** may be formed to surround the upper, side, and lower surfaces of the metal layer **650**A of the extension pattern part **657**.
(220) Accordingly, the lower surface of the first lead pattern part **655** and the upper surface of the spring plate **610** may be spaced apart by a first distance. In addition, the lower surface of the second lead pattern part **656** and the upper surface of the spring plate **610** may be spaced apart by the first distance. In addition, the lower surface of the extension pattern part **657** and the spring plate **610** may be spaced apart by a second distance smaller than the first distance due to the plating layer.
(221) Meanwhile, the thickness of the plating layer **650**B may be 0.3 μm to 1 μm. For example, the thickness of the plating layer **650**B may be 0.3 μm to 0.7 μm. The thickness of the plating layer **650**B may be 0.3 μm to 0.5 μm.
(222) FIG. **15** is a view showing a camera module according to another exemplary embodiment.
(223) Referring to FIG. **15**, the image sensor may be attached to the image sensor substrate **600** in a flip chip bonding method.
(224) That is, after the image sensor in the previous embodiment is attached on the insulating layer **630**, the first lead pattern part **655** and the electrode **710** of the image sensor **700** are mutually connected by wire bonding through a connection member.
(225) Alternatively, the image sensor may be mounted on the package substrate, and accordingly, the image sensor may be mounted on the image sensor substrate **600** on the insulating layer **630** in a flip chip bonding method.
(226) The package substrate **900** may include an insulating layer **910**, a circuit pattern **920**, a via **930**, an image sensor **940**, a connection member **950**, a protection member **960**, and an adhesive member **970**.
(227) The image sensor **940** may be electrically connected to the circuit patterns **920** respectively disposed on the upper and lower surfaces of the insulating layer **910** through the connection member **950**. Further, the via **930** may electrically connect the circuit patterns **920** disposed on the upper and lower surfaces of the insulating layer **910** to each other.
(228) The protection member **960** may be disposed on the lower surface of the insulating layer **910** to protect the lower surface of the insulating layer **910** and expose at least a part of the circuit pattern **920**. In addition, an adhesive member **970** may be disposed on the circuit pattern exposed through the protection member **960**. The adhesive member **970** may be a solder ball. The adhesive member **970** may contain materials of heterogeneous components in solder. The solder may be

composed of at least one of SnCu, SnPb, and SnAgCu. In addition, the material of the heterogeneous component may include any one of Al, Sb, Bi, Cu, Ni, In, Pb, Ag, Sn, Zn, Ga, Cd, and Fe.
(229) Meanwhile, in a state in which the package substrate **900** as described above is manufactured, the adhesive member **970** may be connected to the first lead pattern part **655**.
(230) FIG. **16** is a view showing a camera device according to another exemplary embodiment.
(231) Referring to FIG. **16**, the camera device according to the embodiment may include a lens barrel **1100**, a lens assembly **1200**, a housing **1300**, an infrared cut filter **1400**, driving parts **1510**, **1520**, **1530**, and an image sensor substrate **1600**, an image sensor **1700**, and a flexible circuit board **1800**.
(232) Here, since the lens barrel **1100**, the lens assembly **1200**, the housing **1300**, and the infrared cut filter **1400** have already been described in FIG. **2**, a detailed description thereof will be omitted.
(233) At this time, in FIG. **2**, the image sensor substrate **600** is moved with respect to the lens barrel by the spring plate **610**. Alternatively, in the camera device of FIG. **16**, the spring plate **610** is removed, and an elastic region is formed on the insulating layer **1610** to enable movement of the image sensor substrate **1600**.
(234) The image sensor substrate **1600** is a substrate on which the image sensor **1700** is mounted. Specifically, the image sensor substrate **1600** may be driven by the driving parts **1510**, **1520**, and **1530** to move the image sensor **1700** in the X-axis, Y-axis, and Z-axis directions. Also, the image sensor substrate **1600** may be driven by the driving parts **1510**, **1520**, and **1530** to tilt the image sensor **1700**.
(235) The image sensor substrate **1600** may be disposed to be spaced apart from a bottom surface of the housing **1300** by a predetermined distance. In addition, the image sensor substrate **1600** may relatively move the mounted image sensor **1700** with respect to the housing **1300**.
(236) To this end, the image sensor substrate **1600** may include an insulating layer **1610**, a first bonding sheet **1620**, a conductive pattern portion **1640**, and a support layer **1650**. Also, according to an embodiment, the image sensor substrate **1600** may further include a second bonding sheet **1630**. That is, the second bonding sheet **1630** may be selectively disposed between the insulating layer **1610** and the conductive pattern portion **1640**. When the second bonding sheet is included, a planar area of the second bonding sheet in one embodiment may be same as a planar area of the insulating layer **1610**, and a planar area of the second bonding sheet in another embodiment may be same as a planar area of the conductive pattern portion **1640**.
(237) The insulating layer **1610** may move the image sensor **1700** disposed on the image sensor substrate **1600** in the X-axis, Y-axis, and Z-axis directions while supporting the conductive pattern portion **1640**. To this end, the insulating layer **1610** may include an elastic region having a certain elastic force. Preferably, the insulating layer **1610** may include a plurality of elastic regions. For example, the insulating layer **1610** may include four elastic regions.
(238) That is, at least one region of the insulating layer **1610** may have a certain elastic force, accordingly, while the image sensor substrate **1600** is elastically supported, the image sensor substrate **1600** may be moved in the X-axis, Y-axis, and Z-axis directions.
(239) At this time, the insulating layer **1610** is a substrate for forming the conductive pattern portion **1640**, this may include all of a print, a wiring board, and an insulating substrate made of an insulating material capable of forming the conductive pattern part **1640** on the surface of the insulating layer.
(240) A conductive pattern portion **1640** is disposed on the insulating layer **1610**. The conductive pattern portion **1640** may include lead pattern parts spaced apart from each other by a predetermined distance on the insulating layer **1610**. For example, the conductive pattern portion **1640** may include a first lead pattern part connected to the image sensor **1700** and a second lead pattern part connected to the flexible circuit board **1800**. Also, the conductive pattern portion **1650** may include an extension pattern part connecting between the first lead pattern part and the second

lead pattern part. The first lead pattern part, the second lead pattern part, and the extension pattern part will be described in detail below.
(241) Also, the insulating layer **1610** may include a first insulating part on which the first lead pattern part is disposed, a second insulating part on which the second lead pattern part is disposed, and an extension insulating part on which the extension pattern part is disposed. The first insulating part, the second insulating part, and the extension insulating part will be described in more detail below.
(242) Meanwhile, a bonding sheet **1630** may be disposed between the insulating layer **1610** and the conductive pattern portion **1640**. The bonding sheet **1630** may provide an adhesive force between the insulating layer **1610** and the conductive pattern portion **1640**. In this case, the bonding sheet **1630** may have a shape corresponding to the insulating layer **1610**. To this end, the bonding sheet **1630** may include an elastic region having an elastic force corresponding to the insulating layer **1610**. For example, the bonding sheet **1630** may include a first bonding part, a second bonding part, and an extension bonding part between the first bonding part and the second bonding part.
(243) In the embodiment, the image sensor substrate may move the image sensor by providing an elastic force from the extension insulating part of the insulating layer **1610**, the extension bonding part of the bonding sheet **1630**, and the extension pattern part of the conductive pattern portion **1640**. Preferably, the extension insulating part, the extension bonding part, and the extension pattern part may be referred to as an elastic member. In addition, the elastic member may have a layered structure including an insulating layer region, a bonding layer region, and a pattern layer region. This will be described in more detail below. In addition, the second bonding sheet **1630** may be omitted in the image sensor substrate according to the embodiment. For example, the conductive pattern portion **1640** may be directly disposed on the insulating layer **1610** without the second bonding sheet **1630**. That is, an upper surface of the extension insulating part and a lower surface of the extension pattern part may directly contact each other without the extension bonding part of the insulating layer **1610**. Accordingly, in an embodiment, only the extension insulating part and the extension pattern part may serve as an elastic member without the extension bonding part.
(244) FIG. **17** is a view showing the insulating layer shown in FIG. **16**.
(245) Referring to FIG. **17**, the insulating layer **1610** may elastically support the second bonding sheet **1630** and the conductive pattern portion **1640** constituting the image sensor substrate **1600**.
(246) The insulating layer **1610** may have an elastic region having a certain elastic force to move the image sensor **1700** disposed on the image sensor substrate **1600** in the X-axis, Y-axis, and Z-axis directions.
(247) To this end, the elastic region may include an extension insulating part **1614** having elasticity of the insulating layer **1610**. Specifically, the insulating layer **1610** may have a first insulating part **1611** disposed at the center of the insulating layer **1610**. In addition, a second insulating part **1612** may be disposed to surround the first insulating part **611** at a position spaced apart from the first insulating part **1611** by a predetermined interval.
(248) In addition, the insulating layer **1610** may include a first insulating part **1611** and a second insulating part **1612**, and an open region **1613** may form between the first insulating part **1611** and the second insulating part **1612**. Preferably, the first insulating part **1611** and the second insulating part **1612** may be spaced apart from each other with the open region **1613** therebetween.
(249) In addition, one end of the extension insulating part **1614** may be connected to the first insulating part **1611**, and the other end may be connected to the second insulating part **1612**. To this end, a plurality of extension insulating parts **1614** may be provided. Preferably, the extension insulating part **1614** may include first to fourth extension insulating parts **1614**.
(250) The first insulating part **1611** may include a plurality of outer parts. Preferably, the first insulating part **1611** may include a left outer part, a right outer part, an upper outer part, and a lower outer part.
(251) The second insulating part **1612** may include a plurality of inner parts. Preferably, the second

insulating part **1612** may include a left inner part, a right inner part, an upper inner part, and a lower inner part.

(252) In this case, the left outer part of the first insulating part **1611** and the left inner part of the second insulating part **1612** may be disposed to face each other. In addition, the first extension insulating part constituting the extension insulating part **1614** may connect the left outer part of the first insulating part **1611** and the left inner part of the second insulating part **1612** to each other.

(253) Also, the right outer part of the first insulating part **1611** and the right inner part of the second insulating part **1612** may be disposed to face each other. In addition, the second extension insulating part constituting the extension insulating part **1614** may connect the right outer part of the first insulating part **1611** and the right inner part of the second insulating part **1612** to each other.

(254) In addition, the upper outer part of the first insulating part **1611** and the upper inner part of the second insulating part **1612** may be disposed to face each other. In addition, the third extension insulating part constituting the extension insulating part **1614** may connect the upper outer part of the first insulating part **1611** and the upper inner part of the second insulating part **1612** to each other.

(255) In addition, the lower outer part of the first insulating part **1611** and the lower inner part of the second insulating part **1612** may be disposed to face each other. In addition, the fourth extension insulating part constituting the extension insulating part **1614** may connect the lower outer part of the first insulating part **1611** and the lower inner part of the second insulating part **1612** to each other.

(256) Also, the first insulating part **1611**, the second insulating part **1612**, and the extension insulating part **1614** may be integrally formed. Through this, it is possible to further utilize the elastic force of the insulating layer **1610** when the image sensor is tilted, and it is possible to prevent separation between the first insulating part **1611**, the extension insulating part **1614**, and the second insulating part **1612**.

(257) That is, the insulating layer **1610** may be etched or physically punched to have a spring shape of the extension insulating part **1614** on one insulating member to form the open region **1613**. Accordingly, the first insulating part **1611**, the second insulating part **1612**, and the extension insulating part **1614** may be formed of the same insulating material. However, the embodiment is not limited thereto.

(258) In other words, each of the first insulating part **1611**, the second insulating part **1612**, and the extension insulating part **1614** may be formed in separate configurations. That is, the insulating layer **1610** according to the embodiment may be formed by further forming a configuration corresponding to the extension insulating part **1614** between the first insulating part **1611** and the second insulating part **1612**, after preparing the first insulating part **1611** and the second insulating part **1614**.

(259) On the other hand, the length of the extension insulating part **1614** is set to have at least 1.5 times the linear distance between the first insulating part **1611** and the second insulating part **1612**. In addition, the length of the extension insulating part **1614** is set to be 20 times or less than the linear distance between the first insulating part **1611** and the second insulating part **1614**. Preferably, the length of the extension insulating part **1614** is set to be less than 4 times the linear distance between the first insulating part **1611** and the second insulating part **1612**.

(260) Here, the linear distance may mean a distance between an outer surface and an inner surface facing each other in the first insulating part **1611** and the second insulating part **1612**. Preferably, the linear distance may be a distance between the left outer surface of the first insulating part **1611** and the left inner surface of the second insulating part **1612**. In addition, the linear distance may be a distance between the right outer surface of the first insulating part **1611** and the right inner surface of the second insulating part **1612**. Also, the linear distance may be a distance between an upper outer surface of the first insulating part **1611** and an upper inner surface of the second

insulating part **1612**. Also, the linear distance may be a distance between a lower outer surface of the first insulating part **1611** and a lower inner surface of the second insulating part **1612**.
(261) Meanwhile, a linear distance between the first insulating part **1611** and the second insulating part **1612** may be 1.5 mm. At this time, if less than 1.5 times the linear distance between the first insulating part **1614** and the second insulating part **1612** of the extension insulating part **1614**, the mobility of the image sensor substrate **1600** may be reduced due to the decrease in elastic force of the extension insulating part **1614**. In addition, if the length of the extension insulating part **1614** is greater than 20 times the linear distance between the first insulating part **1614** and the second insulating part **1612**, an image sensor **1700** disposed on the insulating layer **1610** cannot be stably supported, and thus a problem may occur in movement accuracy. Accordingly, in the embodiment, in order to improve mobility, the length of the extension insulating part **1614** is set to be less than 4 times the linear distance between the first insulating part **1611** and the second insulating part **1612**.
(262) Accordingly, the extension insulating part **1614** may be formed to have a plurality of bent spring shapes in the open region **1613**.
(263) FIG. **18** is a view illustrating the second bonding sheet shown in FIG. **16**.
(264) Referring to FIG. **18**, the second bonding sheet **1630** is disposed on the insulating layer **1610**. In this case, the second bonding sheet **1630** has a planar shape corresponding to the planar shape of the insulating layer **1610**.
(265) That is, the second bonding sheet **1630** may include a first bonding part **1631** disposed on the first insulating part **1611** of the insulating layer and a second bonding part **1632** disposed on the second insulating part **1612** of the insulating layer **1610**. In addition, the second bonding sheet **1630** may include an open region **1633** between the first bonding part **1631** and the second bonding part **1632**.
(266) The second bonding part **1632** is disposed to surround the first bonding part **1631** at a position spaced apart from the first bonding part **1631** by a predetermined distance. In this case, the second bonding part **1632** does not directly contact the first bonding part **1631**. Accordingly, the first bonding part **1631** and the second bonding part **1632** may be separated from each other through the open region **1633**.
(267) In addition, the extension bonding part **1634** connecting between the first bonding part **1631** and the second bonding part **1632** is disposed in the open region **1633**. The extension bonding part **1634** has a shape corresponding to the extension insulating part **1614**. The extension bonding part **1634** may be disposed in a region vertically overlapping the extension insulating part **1614**. In this case, the extension bonding part **1634** may have a plane area corresponding to the extension insulating part **1614**. Preferably, the plane area of the extension bonding part **1634** may be 0.9 to 1.1 times the plane area of the extension insulating part **1614**. The extension bonding part **1634** of the second bonding sheet **1630** may be formed through the same process as the extension insulating part **1614** of the insulating layer **1610**. Accordingly, the extension bonding part **1634** may have the same shape as the extension insulating part **1614** and have the same plane area.
(268) The second bonding sheet **1630** may be formed of a double-sided adhesive film. The second bonding sheet **1630** may be formed of an epoxy or acrylic adhesive or a thermosetting adhesive film.
(269) FIG. **19** is a view showing a conductive pattern portion according to an embodiment.
(270) Referring to FIG. **19**, the conductive pattern portion **1640** may be disposed on the insulating layer **1610** with a specific pattern. The conductive pattern portion **1640** includes a first conductive pattern part **1641** disposed on a first region of the insulating layer **1610** and a second conductive pattern part **1642** disposed on a second region of the insulating layer **1610**, a third conductive pattern part **1643** disposed on a third region of the insulating layer **1610**, and a fourth conductive pattern part **1644** disposed on a fourth region of the insulating layer **1610**. In the drawing, a region vertically overlapping the first insulating part **1611** of the insulating layer **1610** and the first bonding part **1631** of the second bonding sheet **1630** is referred to as ‘A’, a region vertically

overlapping the second insulating part **1612** of the insulating layer **1610** and the second bonding part **1632** of the second bonding sheet **1630** is referred to as ‘B’, and a region vertically overlapping the open region **1613** of the insulating layer **1610** and the open region **1633** of the second bonding sheet **1630** is referred to as ‘C’.

(271) Since the structure of the conductive pattern portion shown in FIG. **19** has already been described with reference to FIG. **6**, a detailed description thereof will be omitted.

(272) FIG. **20** is a plan view of a substrate for an image sensor according to another exemplary embodiment.

(273) Referring to FIG. **20**, a second bonding sheet **1630** is disposed on the insulating layer **1610**. In addition, a conductive pattern portion **1640** is disposed on the second bonding sheet **1630**. In this case, an open region is formed in each of the insulating layer **1610** and the second bonding sheet **1630**, and the open regions may overlap each other in a vertical direction.

(274) In addition, an elastic member having an elastic force may be disposed in each of the open regions. Here, the elastic member may refer to a part of the insulating layer **1610** and a part of the second bonding sheet **1630**. That is, the elastic member constitutes a part of the insulating layer **1610** and/or the second bonding sheet **1630**, and thus the insulating layer **1610** and/or the second bonding sheet **1630** may be formed of the same material. That is, an extension insulating part **1614** having a bent shape bent a plurality of times is disposed in the open region of the insulating layer **1610**. The extension insulating part **1614** may have a spring shape. In addition, an extension bonding part **1634** having a bent shape bent a plurality of times is disposed in the open region of the second bonding sheet **1630**. The extension bonding part **1634** may have a spring shape.

(275) Also, the extension bonding part **1634** and the extension insulating part **1614** may have the same shape and the same planar area.

(276) Meanwhile, the line width of the extension insulating part **1614** may be greater than the line width of the extension pattern part **1647**. Accordingly, when the substrate for the image sensor is viewed from the top, at least a part of the extension insulating part **1614** positioned under the extension pattern part **1647** may be exposed.

(277) At this time, the image sensor substrate **1600** not only moves in the X-axis direction and the Y-axis direction, but also moves in the Z-axis direction. At this time, there is a difference between the elastic modulus of the extension insulating part **1614** and the elastic modulus of the extension pattern part **1647**. Accordingly, when the image sensor substrate **1600** moves in the Z-axis direction, the movement distance of the extension insulating part **1614** may appear differently. In addition, this causes a situation in which the extension pattern part **1647** comes into contact with a component made of another metal material, thereby causing an electrical reliability problem (eg, short circuit). Therefore, in the embodiment, as described above, the line width of the extension insulating part **1614** is greater than the line width of the extension pattern part **1647** to solve the electrical reliability problem.

(278) FIG. **21** is a view showing an arrangement structure of a first driving part according to another embodiment.

(279) Referring to FIG. **21**, the first driving part **1510** is disposed on the lower surface of the first insulating part **1611** of the insulating layer **1610**. That is, in FIG. **10**, the first driving part is disposed on the lower surface of the first plate part **611** of the spring plate **610**. Alternatively, referring to FIG. **21**, the first driving part **1510** may be disposed on the lower surface of the first insulating part **1611** of the insulating layer **1610**.

(280) In this case, the first driving part **1510** differs from the first driving part in FIG. **10** only in the arrangement position, but has substantially the same structure and operation characteristics, and thus a detailed description thereof will be omitted.

(281) According to an embodiment, in order to implement the OIS and AF functions of the camera module, instead of moving the conventional lens barrel, the image sensor is moved with respect to the lens barrel in the X-axis, Y-axis and Z-axis directions. Accordingly, the camera module

according to the embodiment may remove a complex spring structure for implementing the OIS and AF functions, and thus the structure may be simplified. In addition, by moving the image sensor according to the embodiment with respect to the lens barrel, it is possible to provide a stable structure compared to the conventional structure.

(282) In addition, according to an embodiment, the extension pattern part electrically connected to the image sensor has a spring structure and is disposed to float on a spring plate. Accordingly, the camera module may move the image sensor while elastically supporting the image sensor more stably. In addition, the length of the extension pattern part is at least 1.5 to 4 times the linear distance between the first lead pattern part and the second lead pattern part. Accordingly, it is possible to minimize noise generation while improving the mobility of the image sensor substrate.

(283) In addition, according to an embodiment, by preventing the elastic member and the extension pattern part from being aligned with each other in the vertical direction, it is possible to solve an electrical reliability problem that may occur due to a contact between the elastic member and the extension pattern part.

(284) In addition, in the insulating layer in the embodiment, an extension insulating part having a spring shape is disposed in a region that vertically overlaps with the extension pattern part, so that the spring plate can be removed. Accordingly, the camera module may move the image sensor with respect to the lens barrel while elastically supporting the image sensor more stably, and may minimize the product volume.

(285) In addition, in the embodiment, the width of the extension insulating part is greater than the width of the extension pattern part, so that the extension pattern part can be stably supported by the extension insulating part, and thus operation reliability can be improved.

(286) In addition, in the embodiment, the first driving part composed of a permanent magnet is disposed so that the same poles are adjacent to each other with respect to the center, while the N poles and the S poles of the permanent magnets adjacent to each other are perpendicular to each other. According to this, in the embodiment, it is possible to secure the arrangement freedom of the second driving part and the third driving part for adjusting the electromagnetic force in response to the permanent magnet, it is possible to move in the X-axis and Y-axis, and stable tilt control is also possible.

(287) Features, structures, effects, and the like described in the embodiments above are included in at least one embodiment, and are not necessarily limited to only one embodiment. Furthermore, the features, structures, effects, and the like illustrated in each embodiment may be combined or modified for other embodiments by a person having ordinary knowledge in the field to which the embodiments belong. Therefore, the contents related to these combinations and modifications should be construed as being included in the scope of the embodiment.

(288) Although the embodiments have been described above, these are only examples and are not intended to limit the embodiments, and those of ordinary skill in the field to which the embodiments belong to various types not exemplified above without departing from the essential characteristics of the present embodiment. It will be seen that branch transformation and application are possible. For example, each component specifically shown in the embodiment can be modified and implemented. And differences related to these modifications and applications should be construed as being included in the scope of the embodiments set in the appended claims.

Claims

1. A substrate for an image sensor comprising: a substrate portion; a conductive pattern portion disposed on the substrate portion; and a first driving part disposed at the substrate portion to move the image sensor in a first axis direction and a second axis direction intersecting the first axis direction, wherein the substrate portion comprises: a spring plate having an elastic member disposed in a first open region, and an insulating layer disposed on the spring plate and having a

second open region overlapping the first open region in a third axis direction perpendicular to the first axis direction and the second axis direction, wherein the spring plate comprises: a first plate part, and a second plate part disposed around the first plate part with the first open region interposed therebetween, and connected to the first plate part by the elastic member, and wherein the first driving part comprises: a first driving part of a first group configured to move the image sensor in the first axis direction, and a first driving part of a second group configured to move the image sensor in the second axis direction.

2. The substrate of claim 1, wherein the insulating layer comprises: a first insulating part disposed on the first plate part, and a second insulating part disposed on the second plate part.

3. The substrate of claim 2, wherein the conductive pattern portion comprises: a first lead pattern part disposed on the first insulating part, and a second lead pattern part disposed on the second insulating part.

4. The substrate of claim 3, wherein the insulating layer further comprises an extension insulating part disposed in the second open region to connect between the first and second insulating parts.

5. The substrate of claim 4, wherein the conductive pattern portion further comprises an extension pattern part disposed on the extension insulating part to connect between the first and second lead pattern parts.

6. The substrate of claim 1, wherein the first driving part is disposed on the first plate part.

7. The substrate of claim 6, wherein the first driving part comprises a coil.

8. The substrate of claim 7, further comprising a second driving part corresponding to the first driving part and including a magnet, wherein the second driving part comprises: a second driving part of a first group corresponding to the first driving part of the first group, and a second driving part of a second group corresponding to the first driving part of the second group.

9. The substrate of claim 8, wherein the first driving part of the first group comprises a first-first driving part and a first-third driving part, wherein the first driving part of the second group comprises a first-second driving part and a first-fourth driving part, wherein the second driving part of the first group comprises a second-first driving part and a second-third driving part corresponding to the first-first driving part and the first-third driving part, and wherein the second driving part of the second group comprises a second-second driving part and a second-fourth driving part corresponding to the first-second driving part and the first-fourth driving part.

10. The substrate of claim 1, wherein the first axis direction is a X axis direction, and wherein the second axis direction is a Y axis direction.

11. The substrate of claim 10, wherein the third axis direction is an optical axis direction, and wherein the X axis direction and the Y axis direction are perpendicular to the optical axis direction.

12. A camera module comprising: a housing; a lens barrel disposed in the housing; a lens assembly disposed in the lens barrel; an image sensor substrate disposed in the housing; first and second driving parts configured to move the image sensor substrate in a first axis direction and a second axis direction intersecting the first axis direction, wherein the first driving part comprises: a first driving part of a first group configured to move the image sensor substrate in the first axis direction with respect to the lens barrel, and a first driving part of a second group configured to move the image sensor substrate in the second axis direction with respect to the lens barrel; wherein the second driving part comprises: a second driving part of a first group overlapping the first driving part of the first group in a third axis direction, and a second driving part of a second group overlapping the first driving part of the second group in the third axis direction.

13. The camera module of claim 12, wherein the third axis direction is an optical axis direction, and wherein the second driving part overlaps with the first-first driving part in the optical axis direction.

14. The camera module of claim 12, wherein the first driving part is disposed at the image sensor substrate, and wherein the second driving part is disposed in a region of the housing facing the first driving part in the third axis direction.

15. The camera module of claim 12, wherein the image sensor substrate includes: a substrate

portion on which an image sensor is placed, and a conductive pattern portion disposed on the substrate portion, wherein the substrate portion includes a spring plate having an elastic member disposed in a first open region, and an insulating layer disposed on the spring plate and having a second open region overlapping the first open area in a third axis direction, and wherein the spring plate includes: a first plate part, and a second plate part disposed around the first plate part with the first open region interposed therebetween and connected to the first plate part through the elastic member.

16. The camera module of claim 15, wherein the insulating layer includes a first insulating part disposed on the first plate part and a second insulating part disposed on the second plate part, and wherein the conductive pattern portion includes a first lead pattern part disposed on the first insulating part and a second lead pattern part disposed on the second insulating part.

17. The camera module of claim 16, wherein the insulating layer further includes an extension insulating part disposed in the second open region and connecting the first insulating part and the second insulating part, and wherein the conductive pattern portion further includes an extension pattern part disposed on the extension insulating part and connecting the first lead pattern part and the second lead pattern part.

18. The camera module of claim 12, wherein the first driving part includes a coil, and wherein the second driving part includes a magnet corresponding to the coil.

19. The camera module of claim 12, wherein the first driving part of the first group comprises a first-first driving part and a first-third driving part, wherein the first driving part of the second group comprises a first-second driving part and a first-fourth driving part, wherein the second driving part of the first group comprises a second-first driving part and a second-third driving part corresponding to the first-first driving part and the first-third driving part, and wherein the second driving part of the second group comprises a second-second driving part and a second-fourth driving part corresponding to the first-second driving part and the first-fourth driving part.